



**Eltrix[®] – Plant Billing and Invoice Settlement System
(PASS) including MDA
for
2000 MW Ar Rass 2 PV Project**

Technical Proposal

Reference: KKK/2324/30798/V1

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1 Executive Summary

1.1 Introduction

This document explains Kalkitech's approach to supply Eltrix Plant Accounting and Settlement System (PASS) software with Meter Data Acquisition (MDA) system for 2000 MW Ar Rass 2 PV project.

1.2 About Kalkitech / ASE

Kalki Communication Technologies Pvt. Limited (Kalkitech) is a technology company founded in 1998 to provide standard-based Control, Communication and Computing solutions to OEMs, System Integrators and End-Users in the energy and utilities industry. We help customers monitor, control, visualize, integrate and optimize their power systems or process systems in real-time. Continuous innovation through research and development is at the core of Kalkitech's corporate philosophy. Over the years, Kalkitech has become the preferred partner for several multinational clients in the energy and utilities industry. Today, Kalkitech's solutions are successfully deployed in over 70 countries, across the full spectrum of the energy value chain. Kalkitech is committed to providing state-of-the-art solutions to the industry, combining the latest advancement in power system engineering and information technology, based on international standards.

1.3 Scope Defined in this Proposal

Scope is to supply the Eltrix PASS and MDA for 2000 MW Ar Rass 2 PV Project. The scope includes design, engineering, testing, commissioning at site and training of PASS software and MDA. Kalkitech has supplied numerous PASS systems in Middle East and India and its credentials are listed in Section 14 and Section 15.

1.4 Deliverables and Acceptance Criteria

S.No.	Item	Type	Acceptance Criteria
1	Functional Design Specification	Document	1. Conform to customer formats 2. Conform to customer requirements.
2	FAT and SAT Procedure	Document	FAT/SAT Procedure is in line with the approved FDS (with respect to “Plant Calculation of Payments compliances”).
3	Internal Test Report	Document	Calculations are as per approved FDS, and results are matching with expected outputs.
4	Engineered Application Software	Software	Approved Test Report
5	User Manual	Document	Delivery Clearance
6	Third Party Documents	Document	Delivery Clearance of the equipment.

2 Proposed Solution

Kalkitech’s proposed solution is based on our proven Eltrix Billing and Settlement software platform, installed at power and water utilities across the world. Monthly invoices for the Project company need to be prepared automatically by billing system in accordance with contractual tariff structure agreement.

For this, the billing system needs to acquire metering data from the fiscal meters, process data from SCADA / DCS system and weather data from weather station to generate the required invoice.

The configuration diagram is as below,

2.1 Solution Architecture:

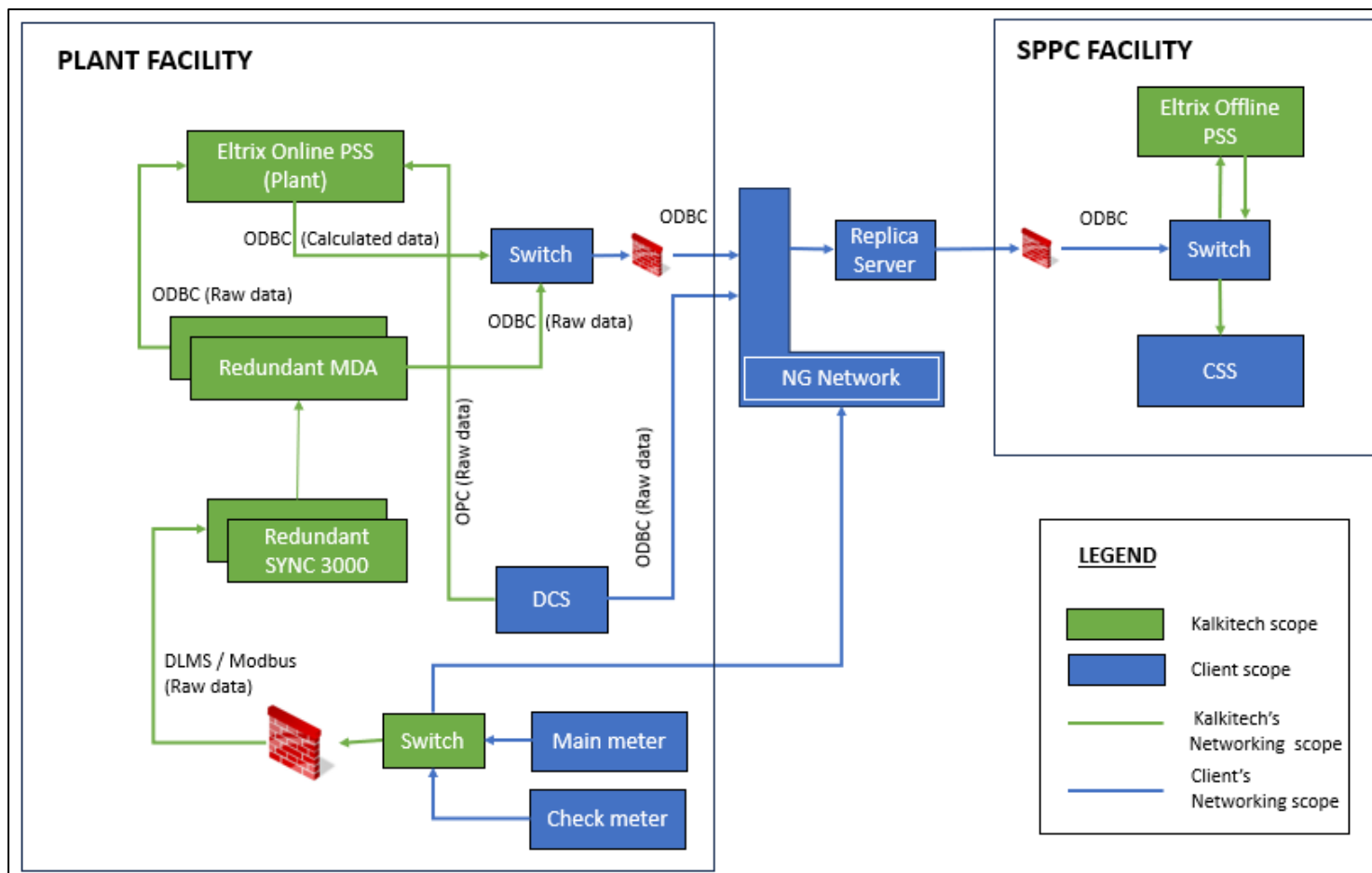


Figure 1- Conceptual diagram for the settlement system

Kalkitech Eltrix Billing System software is a platform for energy and water billing applications, specifically for power and water plants.

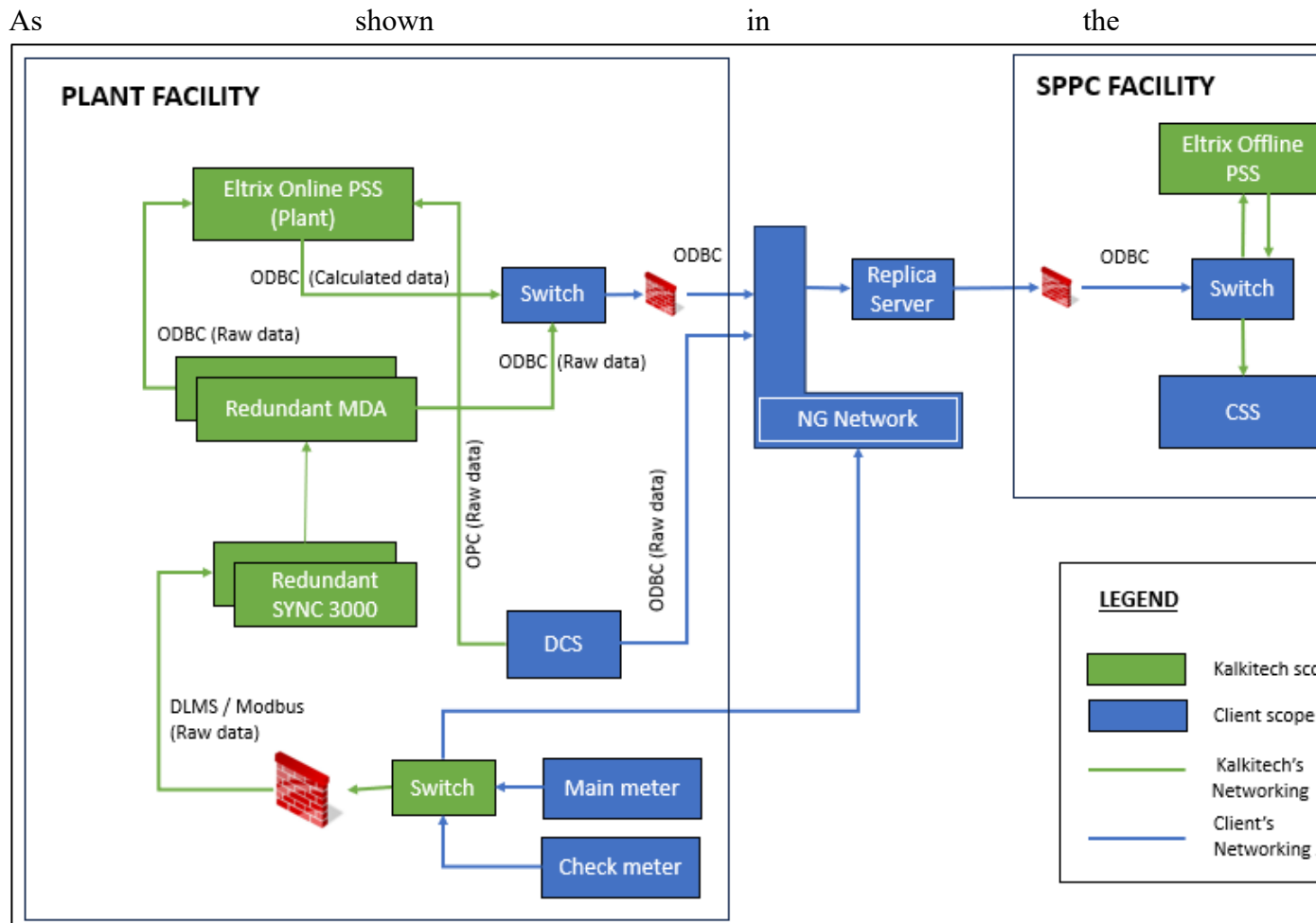


Figure 1- Conceptual diagram for the settlement system. the process and weather-related data from DCS / SCADA shall be made available in ODBC / OPC / Modbus protocol from where Eltrix PASS will fetch the required data. The meter data like energy and irradiance related to PASS shall be made available at switch / firewall in DLMS / Modbus TCP protocol from where Eltrix MDA will fetch the data for computing and generating invoice as per the agreement. The aggregated data is taken by Eltrix MDA software where the data is validated and stored. Eltrix MDA will then push the data to Eltrix PASS software at plant side and replica server via NG network from where data will be pulled by SPPC CSS system and Offline PASS server at SPPC facility.

Similarly, DCS vendor / project company shall make DCS to push the required PASS data to the replica server via NG network from where the data will be pulled by Offline PASS and SPPC CSS system.

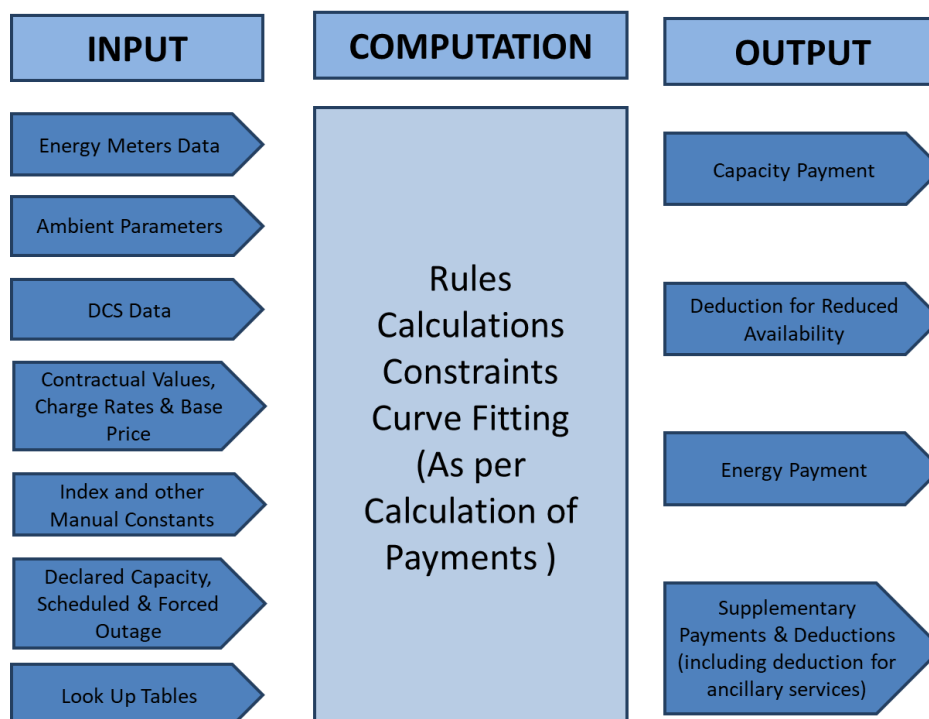


Figure 2- Data Flow Diagram

2.2 Workflow

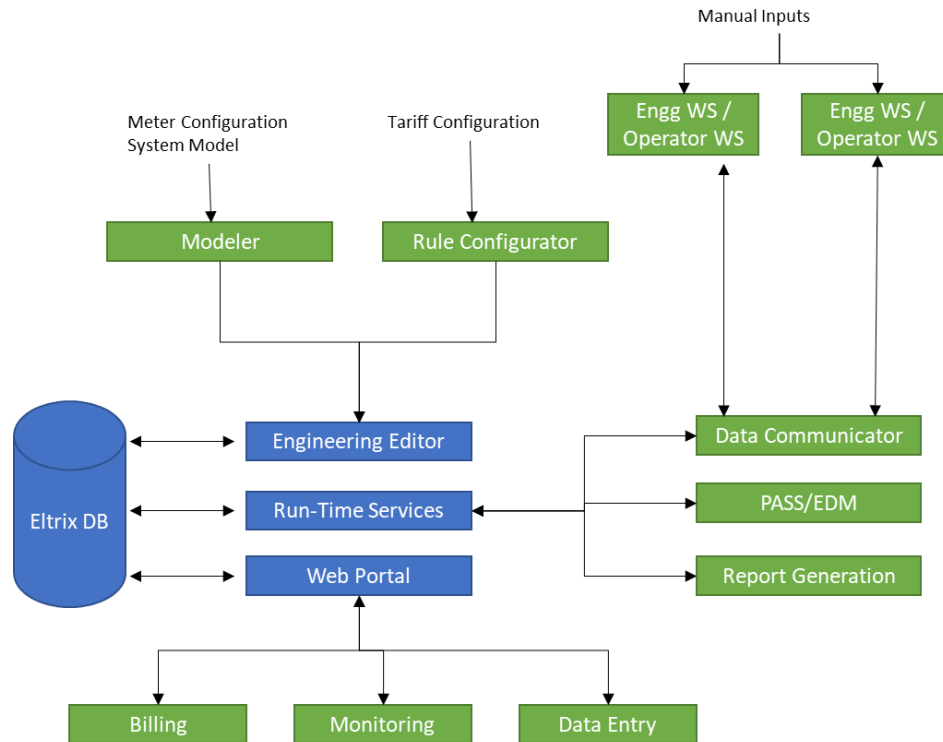


Figure 3 – Workflow

2.3 System Functionality

The main functions of billing system are:

- Meter Data Acquisition System. Basically, it is meter data acquisition for Power output. Data from the energy meter and irradiance meter shall be available in switch / firewall from which Eltrix MDA will fetch the data for PASS solution.
- Required process and weather station data shall be made available in DCS in ODBC / OPC / Modbus from which Eltrix PASS will fetch PASS related data.
- Push the required MDA data to the Replica server using NG network, from where Offline PASS and SPPC CSS fetches the data.
- Tariff calculation formula for implementation of both Online and Offline PASS.
- Invoice Generation as per tariff structure.
- Graphical user interface with customized dashboards.

3 Meter Data Acquisition System

The Meter Data Acquisition system acts as data server for PASS. It basically fetches all the required data from the different data sources like energy meters, irradiance meters, weather station and process data using MDA device interface (SYNC 3000) and store the data at MDA database storage from where the data will be pushed to the Eltrix PASS system and the NG network.

The MDA functionality is split into two,

- A) Device interface, to collect data from device and systems on open protocols like MODBUS, DLMS, IEC104, DNP, OPC, etc.
- B) Database storage to store and process data.

Traditional configuration of MDA combines the above two functionality in a server. The problem with this configuration is that server OS and its hardware gets obsolete very soon, but field devices last at least 15-20 years. This causes a challenge in obsolescence management.

In the above-mentioned approach, the device interface is a hardened industrial interface (SYNC3000) whose job is to communicate on all and any open device protocol and acquire the data.

The acquired data is then stored in the MDA database. So, even if the database and OS gets obsolete, the new system can easily get integrated into SYNC3000 and the system is up and running. If a device becomes obsolete or needs to be changed, any device on any protocol can be integrated into SYNC3000 and thus MDA as a system can be maintained. Further the product for device interface (SYNC3000) that Kalki provides have all type of interface (Serial, ethernet, FO), all protocols with security compliance and will acts as universal hub for all present and future devices.

3.1 MDA – Device interface (SYNC 3000)

Please refer Figure 1- Proposed architecture solution, in page 8, for the meter data acquisition philosophy. We assume Irradiance meter and Energy meters data communicates on MODBUS / DLMS protocols and on ethernet. These meter data get terminated over ethernet on a switch from the metering panel. The MDA device interface - Sync 3000 DCU takes data from these devices on multiple protocols through a switch from the metering panel. Sync3000 DCU then sends the data to redundant Eltrix meter data acquisition data storage server on ODBC / OPC / DLMS protocol.

Similarly, it collects PASS related data from SCADA / DCS and Lab results on OPC / ODBC / Modbus, then sends the data to redundant Eltrix meter data acquisition data storage server.

If there are multiple metering panels, Sync 3000 DCU from all these panels will send to redundant Eltrix meter data acquisition server.

3.1.1. SYNC 3000 – Data Concentrator Unit (DCU)

SYNC 3000 is a high-performance data concentrator and communication device complying to the requirements of IEC 61850-3 for environmental, mechanical, electromagnetic immunity/interferences, that serves as an ideal substation communication hub with advanced features including redundancy, robustness and security. The SYNC 3000 series of products are rugged and come with reliable communication and computing processors capable of handling extreme substation rugged environments. SYNC 3000 DCUs are high-performance front-end gateways for heavy duty substation data concentration and other similar applications. These devices are field proven and secure to meet wide ranging protocol communication integration and translation requirements for large sized applications in power generation, transmission and distribution including Disturbance Data Collection and Wide Area Monitoring. SYNC 3000 supports up to 8 Serial and 3 Ethernet ports, and scalability to support a large number IEDs and other field devices.



Figure 4: SYNC 3000 Physical Appearance

3.1.1.1 SYNC 3000 Hardware Features

The key features of SYNC 3000 product series include:

- CPU - QorIQ integrated quad core communication processor on 1.2 GHz
- 1GB RAM and 4GB onboard flash memory
- Ethernet communication ports - Two 10/100 Mbps ports & four 10/100/1000 Mbps ports with Auto MDIX capability. 100 Mbps Fibre optic ports are also available as a hardware variant.
- Serial communication ports - up to 8 ports supporting wide range of baud-rates
- Time synchronization using IRIG-B DCLS signal Input
- SATA hard disk connectors for storage expansion
- Option for expanding communication ports using terminal server and LAN switches
- Full MODEM port available for connecting external GPRS/CDMA/RF/PSTN MODEM
- Wide range of AC and DC power supply
- Device redundancy – Hot-Standby operation
- Power supply redundancy option
- LEDs for monitoring of power module status, LAN ports link/activity & Serial Tx/Rx
- IEC 61850-3 certified hardware
- Fan-less design
- 19-inch rack mountable, 1U height

The SYNC 3000 station data concentrator has direct Ethernet/Serial connection to field devices. SYNC 3000 has 8 serial ports and 3 Ethernet ports.

3.1.1.2 SYNC 3000 Functions and Features

- Protocol Conversion Engine with extensive list of licensable protocols
- APIs for adding custom protocol and applications
- Multi master communication capability
- Automatic start-up, initialization with restart notification following power restoration
- Disturbance and fault record collection and management
- Time synchronization using IEC 60870/DNP3/SNTP/IRIG-B/IEC 61588/IEEE 1588 PTP version 2.0
- Supports redundant time sync from SNTP servers
- Transparent/tunnelling support for remote management of plant/substation devices
- SNMP Agent/ Manager for NMS Integration
- File upload/download support
- Obtain and store event records and disturbance records from substation devices
- Network tunnelling and routing support
- Built-in Web HMI for monitoring and control
- Syslog support for device logging

3.1.1.3 SYNC 3000 Protocol Support

There are a variety of protocols which are supported and can be licensed in SYNC 3000. The common and important protocols are listed below.

- System Protocols - TCP/IP, UDP/IP, HTTPS, SNMP, ICMP, ARP, SSL, SFTP, SSH
- Standard Protocols - IEC 60870-5-101/103/104, DNP3 serial/TCP, Modbus RTU/ASCII/TCP, IEC 62056-DLMS, IEC 61850, IEC 61400, ICCP, IEEE37.118
- Proprietary Protocols - ABB - RP570, 571, SPA bus, SEL - SEL451/421/311/300G, Schneider - SEPAM Modbus, Areva – Courier, RTK, EXCOM, CMC Master

In addition to the above list, a variety of other standard and proprietary protocols are supported in SYNC 3000. In this proposal, protocol types DLMS and Modbus are considered in addition to SFTP and SNTP.

SYNC 3000 communicates concurrently on all ports. The protocol on which a port communicates can be configured on site from the licensed list of protocols. All status and configuration information which is to be retained during power failure is stored in non-volatile memory.

3.1.1.4 SYNC 3000 Security

- Compliant with NERC-CIP v5
- Compliance with IEEE1686
- Role based access based on OCSP or IEC 62351-8
- Terminal access with two factor authentications
- Inbuilt firewall
- IEC 62351-3 transport layer security
- IEC 62351-5/DNP3 secure authentication
- SSL based VPN support
- Secure password-based locking of configuration project

3.1.1.5 SYNC 3000 – Redundancy

SYNC 3000 supports physical operational redundancy. The redundancy mechanism consists of a hot and warm device. The hot system communicates with the field devices and the upstream system, as if it is a standalone system. The hot and warm system communicates with each other through an integrated link. Integrated link support includes serial, TCP, TCP-serial and TCP-TCP. Multiple modes of switching mechanisms are supported by SYNC 3000. Details can be found in the enclosed document on ‘Redundancy Configuration’.

3.2 MDA – Database Storage

The data from SYNC 3000 will be stored in the MDA database storage and will be aggregate data to redundant Eltrix Plant PASS on OPC and also send data independently to Eltrix PASS system in SPPC / SEC’s HQ facility and Procurer’s Disaster Recovery facility.

4 Eltrix – Billing System Overview

The Eltrix Billing system consists of different software modules. The primary objective is to automatically generate the monthly invoices based on the Power Purchase Agreement (PPA). The real time data required for monthly invoice generation are mainly flow and measurement from Meter Data Server. Apart from real-time data other information manual data like Availability Declaration, Startup information, etc. as per PPA may need to be entered manually. Eltrix Billing system provides platform for configuration, running and user interface for the billing system. Its core components Engineering Editor, Run-Time Services and Web Portal help develop the solution as explained further in the proposal. The Billing System will use a database management system internally to store configuration and measurement data. The system is completely automatic and is not excel based, which make it secure and reliable.

4.1 Billing Model

Eltrix Billing Model calculates various components of the energy charges according to the applicable tariff rules specified in the PPA (Schedule 9 and Schedule 17). The billing model provides a powerful and flexible rule configuration engine, using which users can configure the tariff rules, without requiring any specialized software development expertise. The rules define the various components of the payment, and how they are to be computed as per the PPA. Based on the meter data as well as the plant real-time process data available in the Eltrix database, the billing model computes the invoice tariff payment and stores the results in the database for further processing by the invoicing and settlement system. The calculations will be triggered automatically on a specified time interval or manually on user request. Manual inputs to invoices, if any, will be provided via the user interface and stored in the database.

4.2 Eltrix Engineering Editor

The following components of engineering editors are mainly used for configuring the billing system.

Modeler

Modeler is used for configuring plant model with the sent out electrical lines. The modeler provides the configuration of user security for accessing the system as well. A sample screen shot is given below.

Modeler helps modeling the plant that can predict output for the current operating condition.

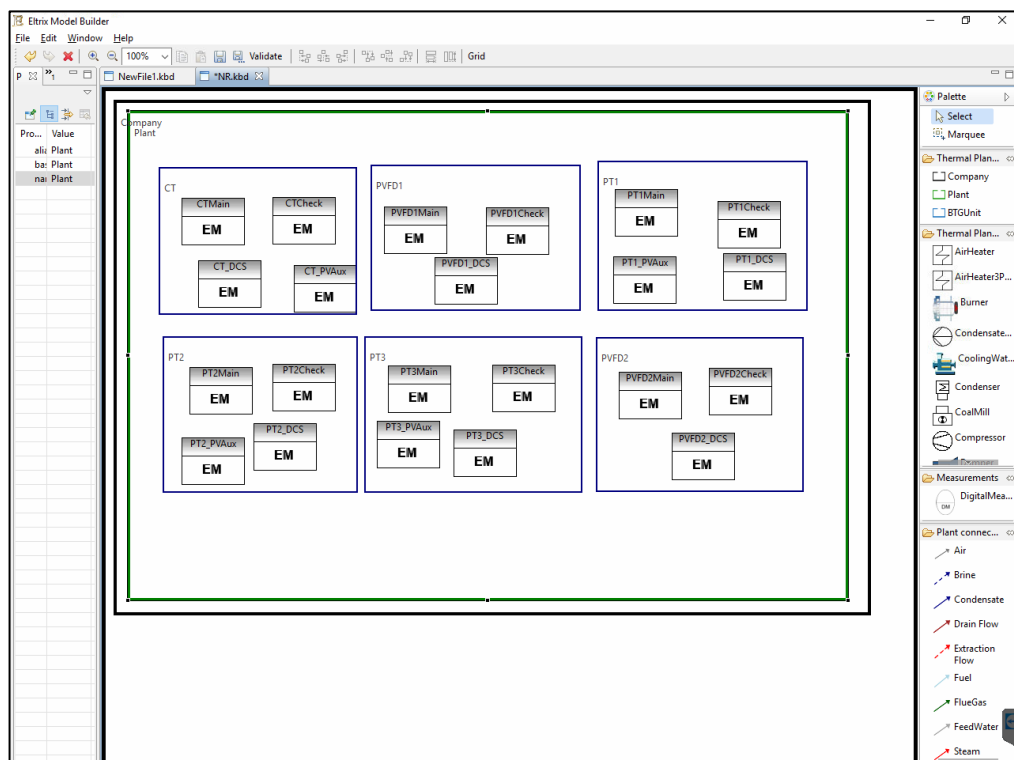


Figure 5- Model Builder

Rule Configuration

Rule configuration is used to configure the tariff calculation procedures as per the PPA contract. The calculation includes Capacity payment, Output payments, Supplemental payment, Penalties, incentives, etc. The rule configuration involves configuration of inputs, such as recorded monthly energy data from tariff meters and then configuration of rule, formula as provided in IPP Calculation of Payments and finally the output data such as monthly invoice amount, etc. A **sample** screen shot is given below, in which Available Capacity calculation has been incorporated.

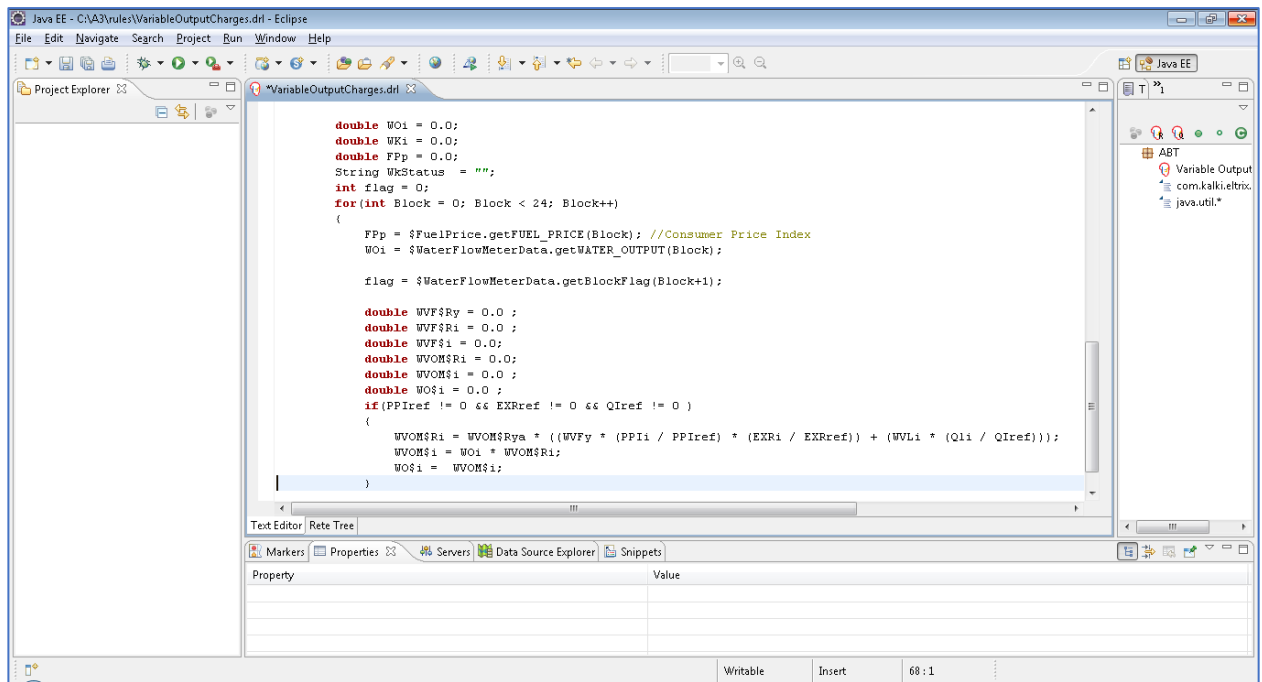


Figure 6- Rules Configurator

4.3 Eltrix – Run-time Services

4.3.1. Data Communicator

Eltrix connects with the meter data server on DLMS or ODBC. Eltrix also has an in-built Flat File Interface which connects to the text files generated by the Kalki Meter Data Gathering driver and fetch meter data for the Billing system to execute invoice calculation. Eltrix – Communicator interfaces include OPC Client and DLMS for two-way data exchange between Meter Data Server and Eltrix PASS server.

4.3.2. Curve Fitting

Curve fitting sub module has the feature to plot/modify standard design curves and correction curves used for calculating the expected standard conditions required for performance comparisons, design value substitutions etc. Using this wizard various degrees of polynomial equation can be generated as per the required accuracy.

Polynomial Equation:

Curve fitting module takes the X, Y values of input tag and curve order to generate a best fit polynomial of n^{th} order. The Gauss elimination method is used to generate the n^{th} order polynomial equation. A good quality tag will be further replaced with the X values in the generated polynomial equation.

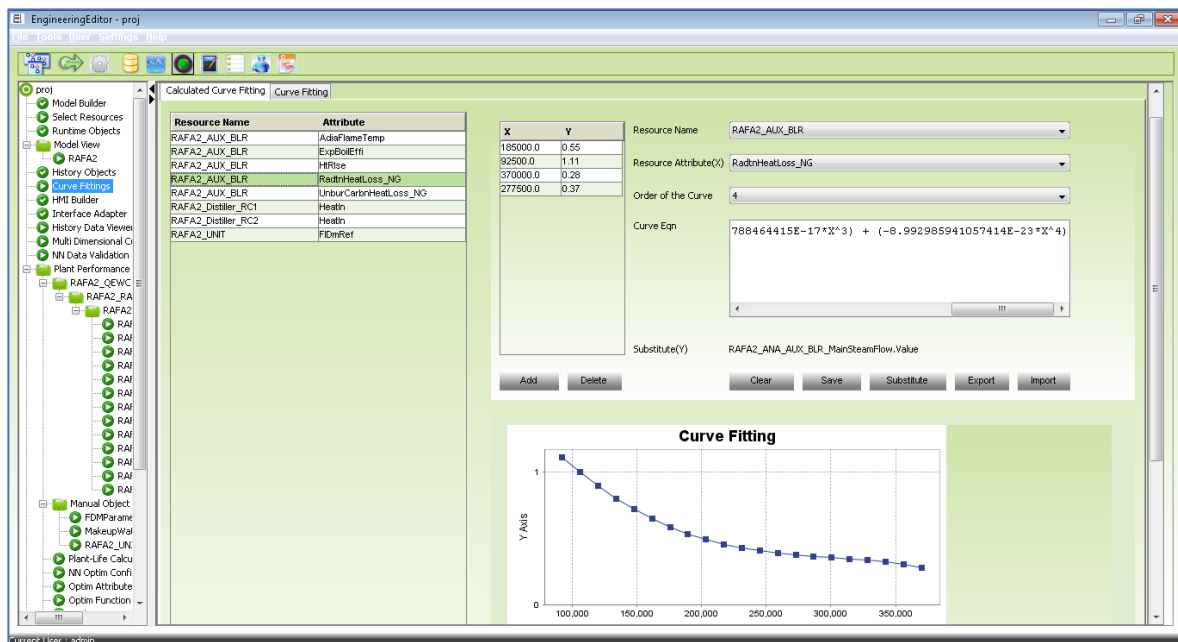


Figure 7- Curve Fitting

A typical polynomial expression of n^{th} order is given below:

$$P(x) = P_1 x^n + P_2 x^{n-1} + \dots + P_n x + P_{n-1}$$

Where x refers to the substitution tag value and P(x) refers to input tag value.

4.3.3. Calculation Builder

This module performs all types of calculations required for estimating the energy charges taking all configured inputs. Calculations are performed continuously and to provide outputs at pre-defined intervals or whenever operator demands. Calculation procedures are defined using Rule configurator. All outputs of the calculation are stored in the database.

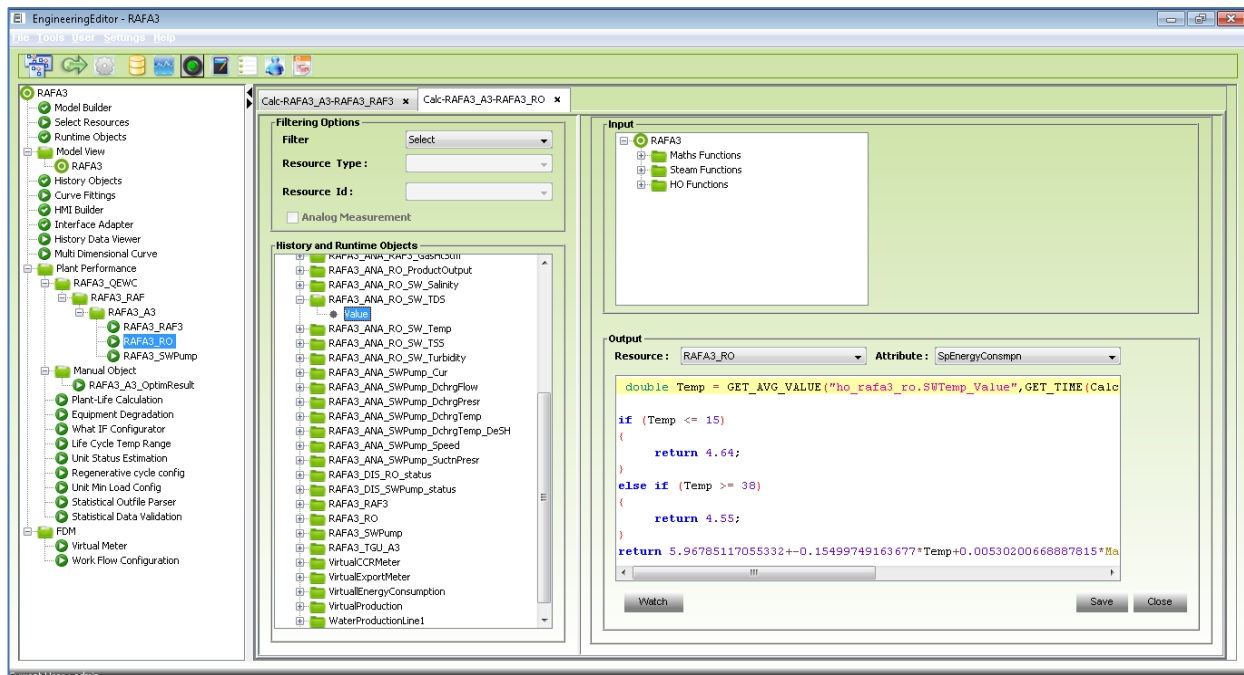


Figure 8- Calculation Builder

4.4 Eltrix – Report Generator

Report Generator is a comprehensive tool for creating and managing reports. Following are the features of report generator:

- Easily create new reports using an intuitive web-based drag and drop Ad Hoc reporting interface.
- Efficiently and securely manage many reports.
- Interact with reports, including entering parameters and drilling on data.
- Arrange reports and web content to create appealing, data-rich dashboards that quickly convey business trends.

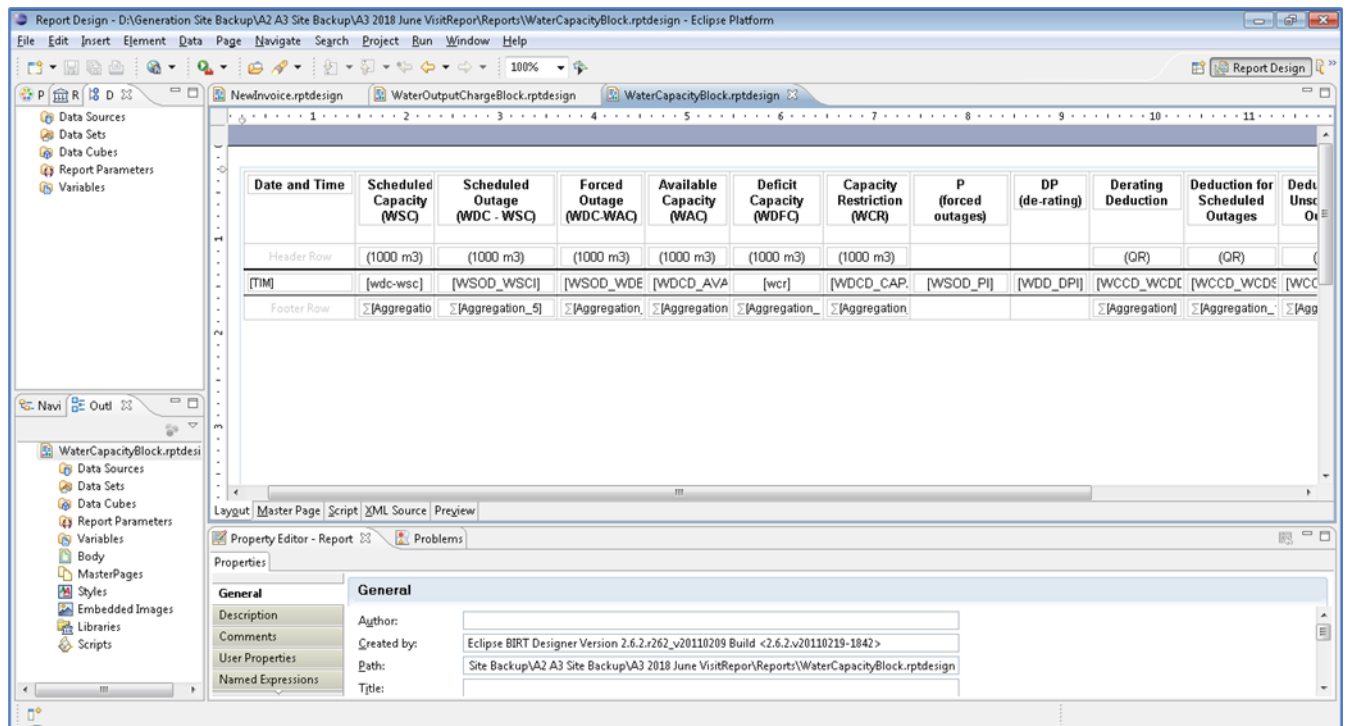


Figure 9- Report Generator

4.5 Eltrix – Web Portal - Monitoring

Graphical displays shall be configured using Eltrix Web Portal to view from web based clients. The system provides a highly flexible display creation facility for creating customized and personalized displays which contain different sets of data available in the system database. Also reports can be viewed on user request. Typical displays which are usually configured for similar applications are the following:

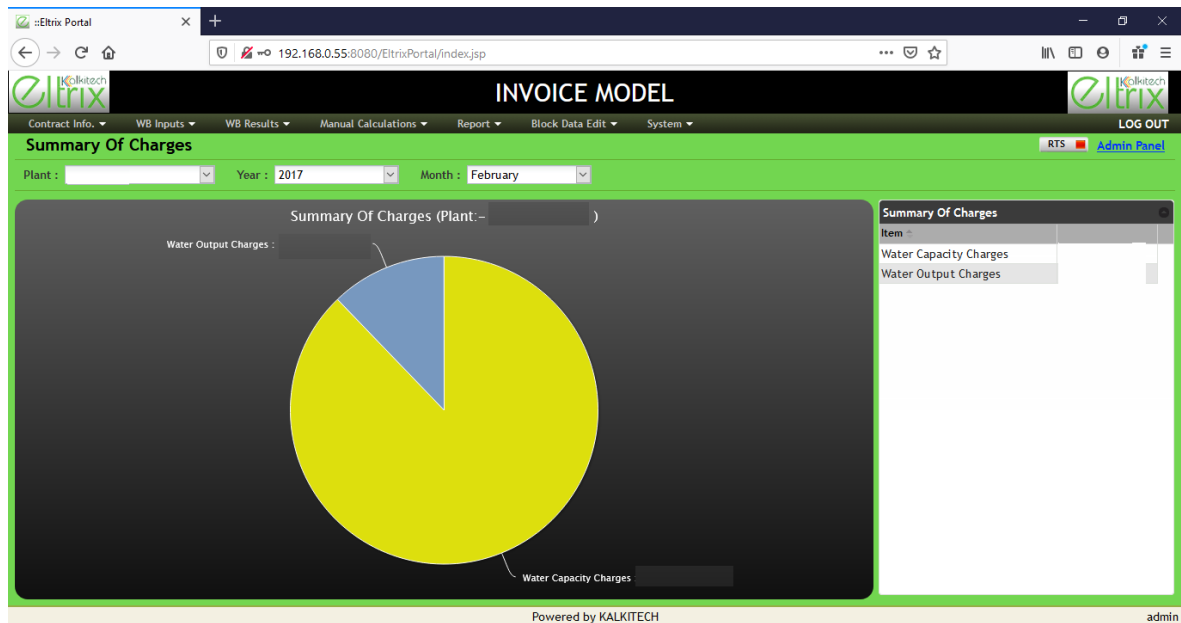
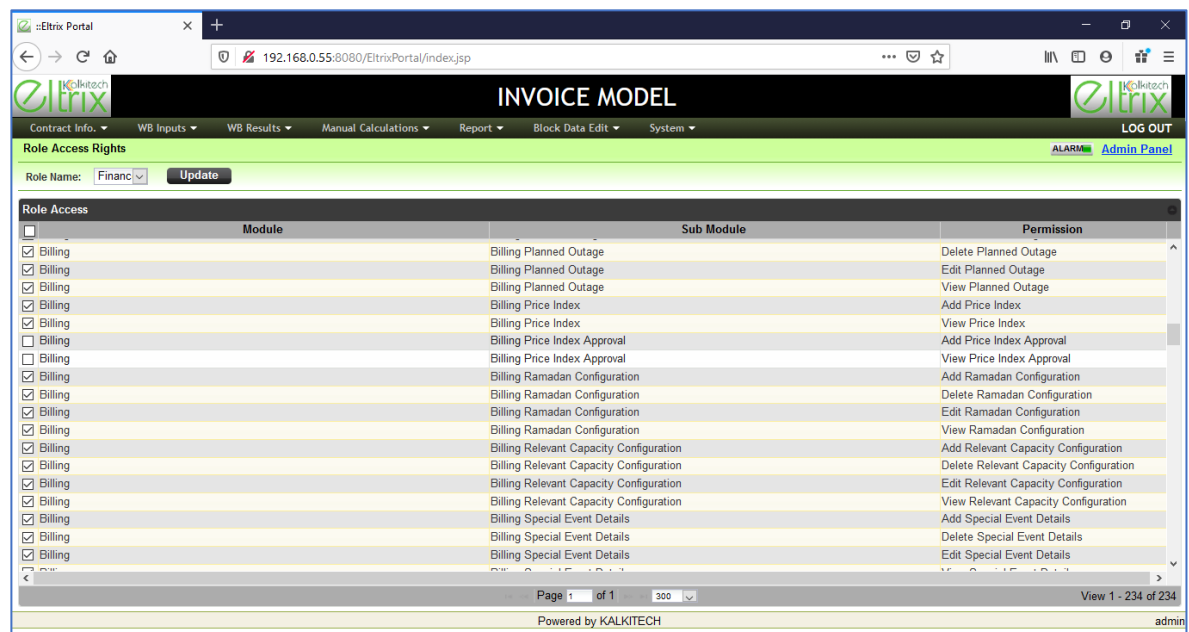
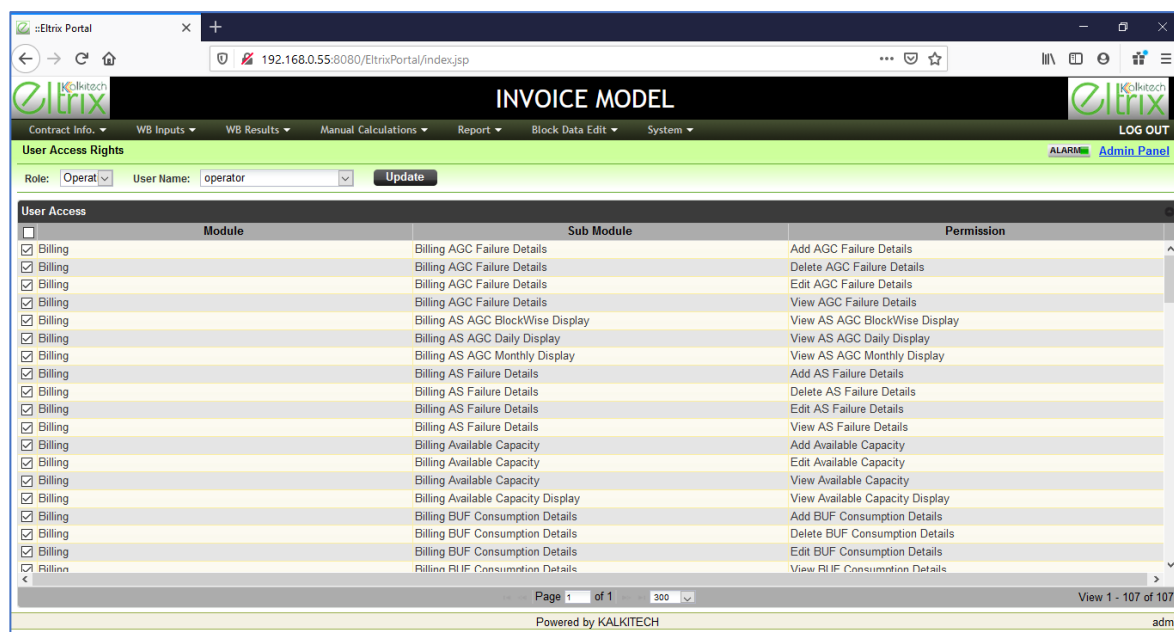


Figure 10- Summary Charges Display



Module	Sub Module	Permission
Billing	Billing Planned Outage	Delete Planned Outage
Billing	Billing Planned Outage	Edit Planned Outage
Billing	Billing Planned Outage	View Planned Outage
Billing	Billing Price Index	Add Price Index
Billing	Billing Price Index	View Price Index
Billing	Billing Price Index Approval	Add Price Index Approval
Billing	Billing Price Index Approval	View Price Index Approval
Billing	Billing Ramadan Configuration	Add Ramadan Configuration
Billing	Billing Ramadan Configuration	Delete Ramadan Configuration
Billing	Billing Ramadan Configuration	Edit Ramadan Configuration
Billing	Billing Ramadan Configuration	View Ramadan Configuration
Billing	Billing Relevant Capacity Configuration	Add Relevant Capacity Configuration
Billing	Billing Relevant Capacity Configuration	Delete Relevant Capacity Configuration
Billing	Billing Relevant Capacity Configuration	Edit Relevant Capacity Configuration
Billing	Billing Relevant Capacity Configuration	View Relevant Capacity Configuration
Billing	Billing Special Event Details	Add Special Event Details
Billing	Billing Special Event Details	Delete Special Event Details
Billing	Billing Special Event Details	Edit Special Event Details

Figure 11- Role Access Rights Configuration Screen



INVOICE MODEL

Contract Info. ▾ WB Inputs ▾ WB Results ▾ Manual Calculations ▾ Report ▾ Block Data Edit ▾ System ▾ [LOG OUT](#)

User Access Rights [ALARMS](#) [Admin Panel](#)

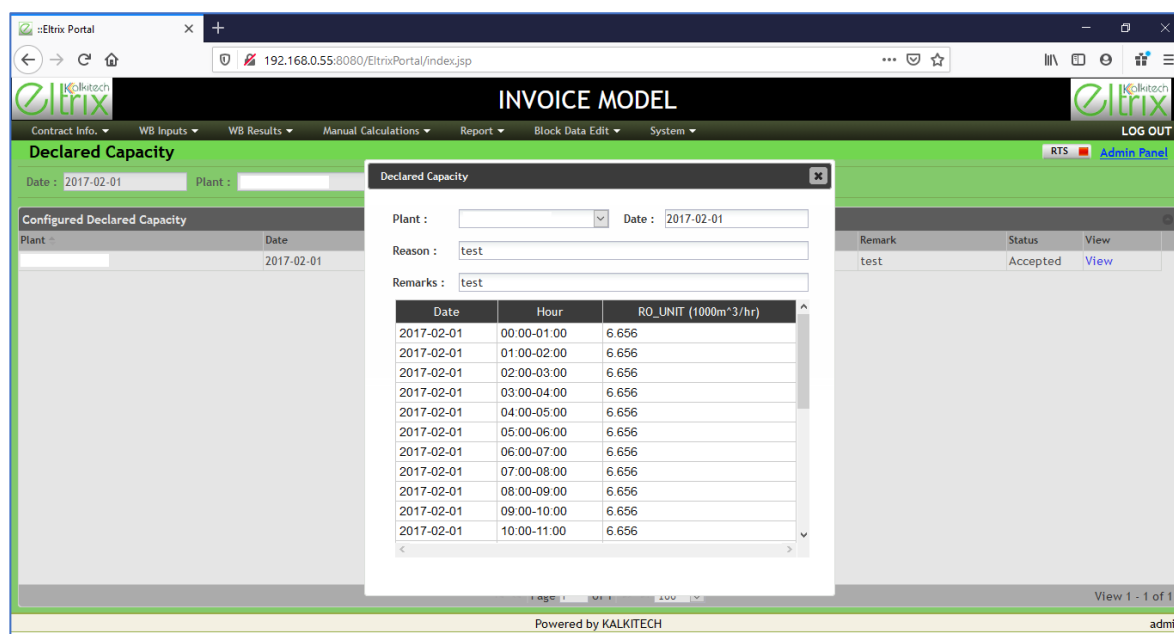
Role: **Operator** User Name: **operator** [Update](#)

Module	Sub Module	Permission
☒ Billing	Billing AGC Failure Details	Add AGC Failure Details
☒ Billing	Billing AGC Failure Details	Delete AGC Failure Details
☒ Billing	Billing AGC Failure Details	Edit AGC Failure Details
☒ Billing	Billing AGC Failure Details	View AGC Failure Details
☒ Billing	Billing AS AGC BlockWise Display	View AS AGC BlockWise Display
☒ Billing	Billing AS AGC Daily Display	View AS AGC Daily Display
☒ Billing	Billing AS AGC Monthly Display	View AS AGC Monthly Display
☒ Billing	Billing AS Failure Details	Add AS Failure Details
☒ Billing	Billing AS Failure Details	Delete AS Failure Details
☒ Billing	Billing AS Failure Details	Edit AS Failure Details
☒ Billing	Billing AS Failure Details	View AS Failure Details
☒ Billing	Billing Available Capacity	Add Available Capacity
☒ Billing	Billing Available Capacity	Edit Available Capacity
☒ Billing	Billing Available Capacity	View Available Capacity
☒ Billing	Billing Available Capacity Display	View Available Capacity Display
☒ Billing	Billing BUF Consumption Details	Add BUF Consumption Details
☒ Billing	Billing BUF Consumption Details	Delete BUF Consumption Details
☒ Billing	Billing BUF Consumption Details	Edit BUF Consumption Details
☒ Billing	Billing BUF Consumption Details	View BUF Consumption Details

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Powered by KALKITECH admin

Figure 12- User Access Rights Configuration Screen



INVOICE MODEL

Contract Info. ▾ WB Inputs ▾ WB Results ▾ Manual Calculations ▾ Report ▾ Block Data Edit ▾ System ▾ [LOG OUT](#)

Declared Capacity [RTS](#) [Admin Panel](#)

Date: **2017-02-01** Plant:

Configured Declared Capacity

Plant	Date
	2017-02-01

Declared Capacity

Plant: Date: **2017-02-01**

Reason: **test**

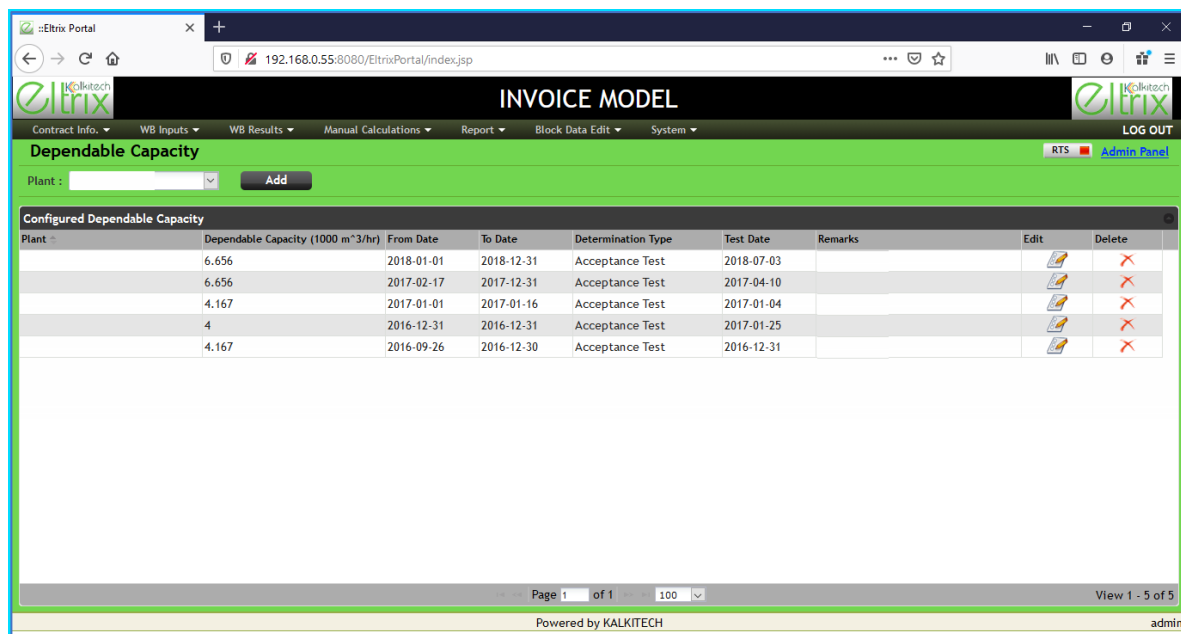
Remarks: **test**

Date	Hour	RO_UNIT (1000m ³ /hr)
2017-02-01	00:00-01:00	6.656
2017-02-01	01:00-02:00	6.656
2017-02-01	02:00-03:00	6.656
2017-02-01	03:00-04:00	6.656
2017-02-01	04:00-05:00	6.656
2017-02-01	05:00-06:00	6.656
2017-02-01	06:00-07:00	6.656
2017-02-01	07:00-08:00	6.656
2017-02-01	08:00-09:00	6.656
2017-02-01	09:00-10:00	6.656
2017-02-01	10:00-11:00	6.656

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Figure 13- Declared Capacity Configuration Screen



Dependable Capacity

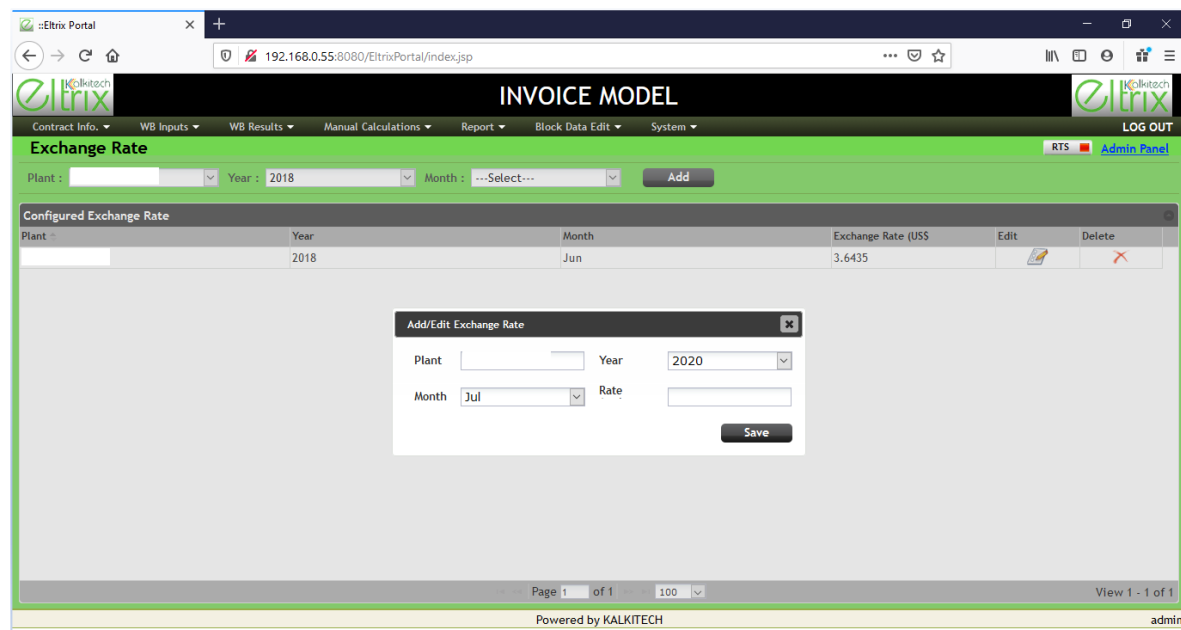
Plant : Add

Plant	Dependable Capacity (1000 m ³ /hr)	From Date	To Date	Determination Type	Test Date	Remarks	Edit	Delete
	6.656	2018-01-01	2018-12-31	Acceptance Test	2018-07-03			
	6.656	2017-02-17	2017-12-31	Acceptance Test	2017-04-10			
	4.167	2017-01-01	2017-01-16	Acceptance Test	2017-01-04			
	4	2016-12-31	2016-12-31	Acceptance Test	2017-01-25			
	4.167	2016-09-26	2016-12-30	Acceptance Test	2016-12-31			

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Figure 14- Dependable Capacity Configuration Screen



Exchange Rate

Plant : Year : 2018 Month : ---Select--- Add

Plant	Year	Month	Exchange Rate (US\$)	Edit	Delete
	2018	Jun	3.6435		

Add/Edit Exchange Rate

Plant Year

Month Rate

Save

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Figure 15- Exchange Rate Configuration Screen

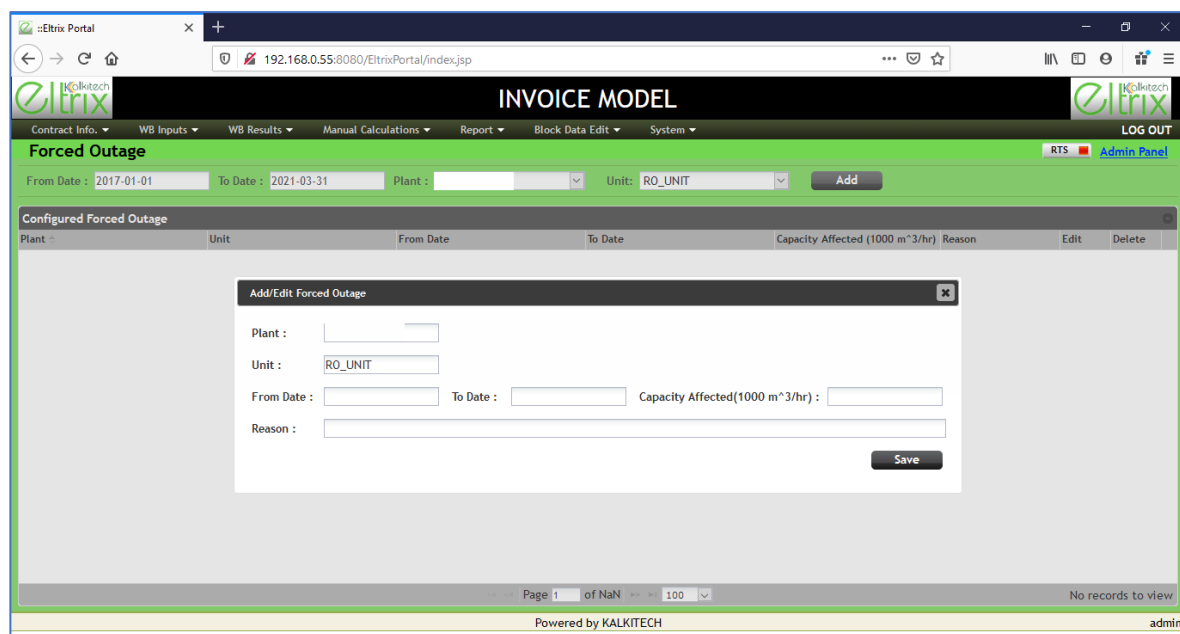


Figure 16- Forced Outage Configuration Screen

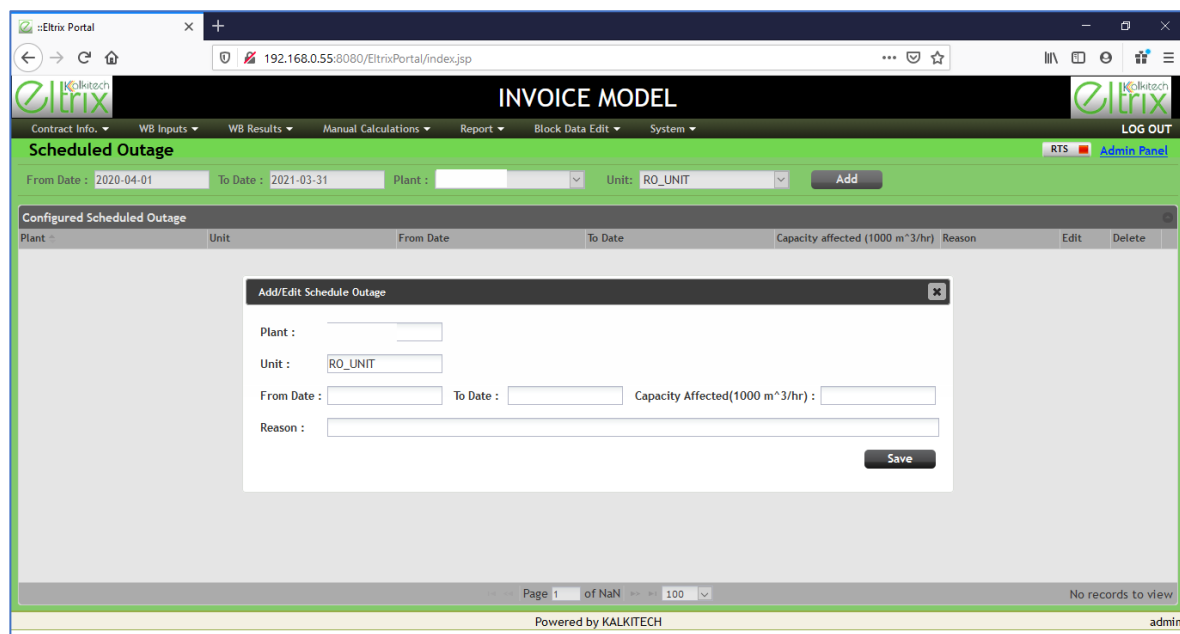
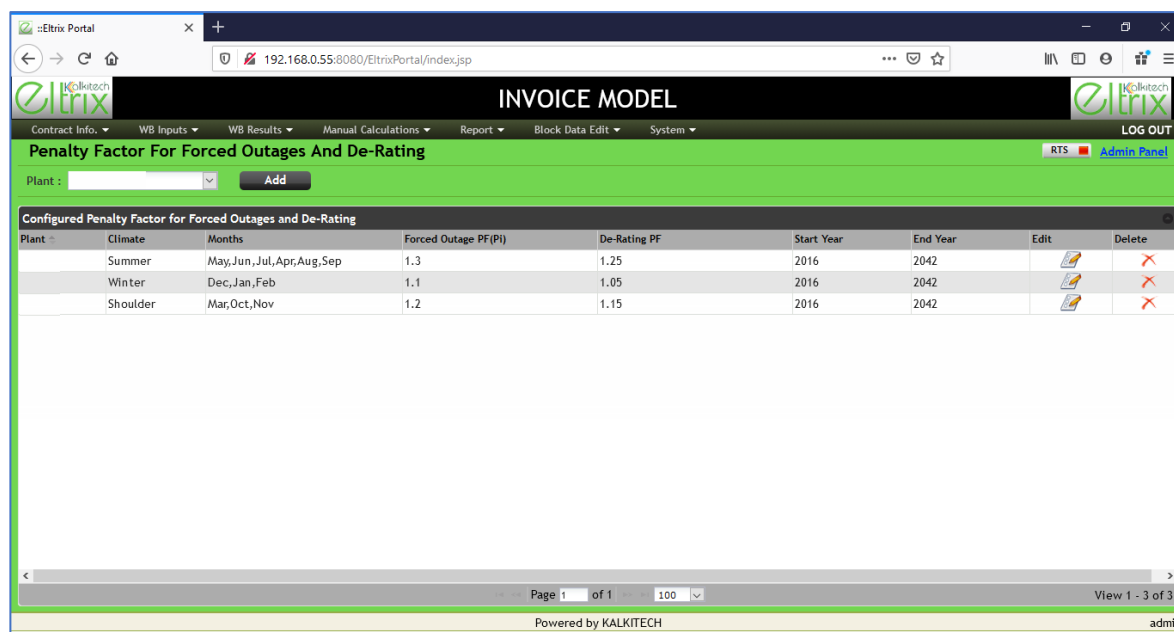


Figure 17- Scheduled Outage Configuration Screen

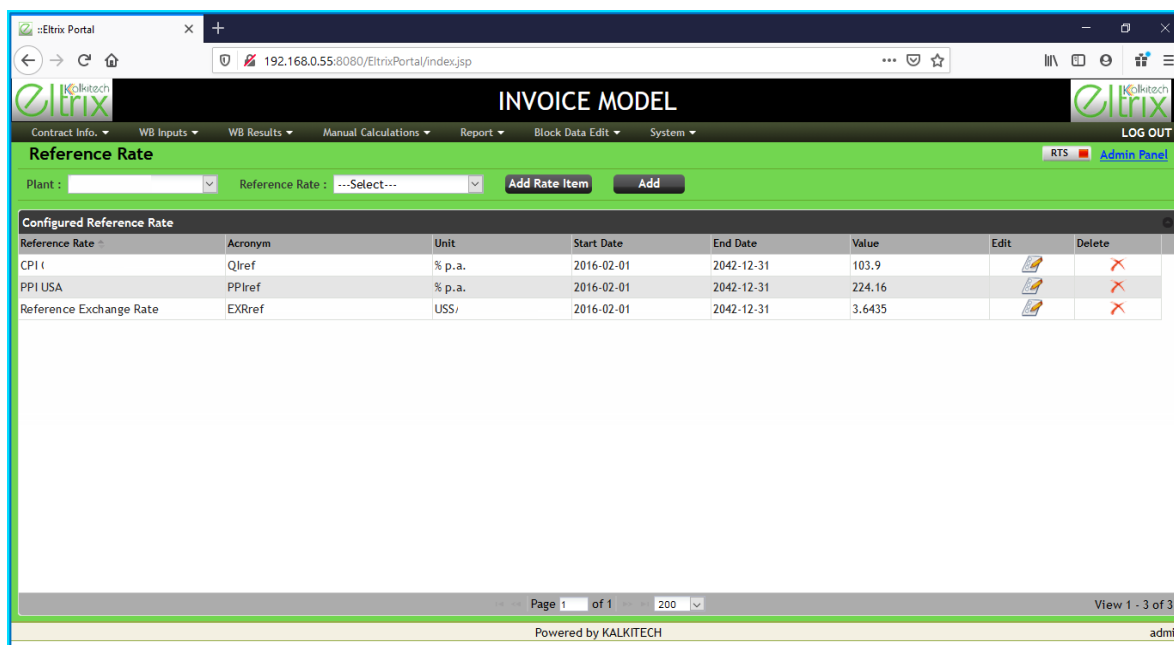


Penalty Factor For Forced Outages And De-Rating

Plant	Climate	Months	Forced Outage PF(Pi)	De-Rating PF	Start Year	End Year	Edit	Delete
	Summer	May, Jun, Jul, Apr, Aug, Sep	1.3	1.25	2016	2042		
	Winter	Dec, Jan, Feb	1.1	1.05	2016	2042		
	Shoulder	Mar, Oct, Nov	1.2	1.15	2016	2042		

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Figure 18- Penalty Factor/Deduction Factor Configuration Screen

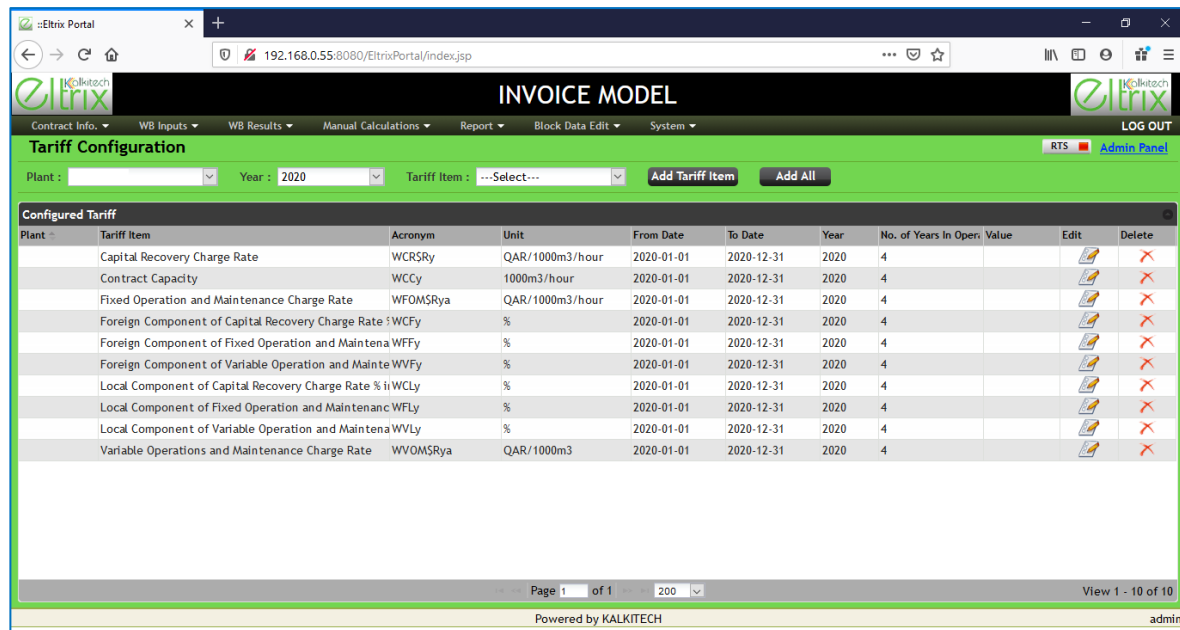


Reference Rate

Reference Rate	Acronym	Unit	Start Date	End Date	Value	Edit	Delete
CPI	Qlref	% p.a.	2016-02-01	2042-12-31	103.9		
PPI USA	PPlref	% p.a.	2016-02-01	2042-12-31	224.16		
Reference Exchange Rate	EXRref	US\$/	2016-02-01	2042-12-31	3.6435		

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Figure 19- Reference Rate Configuration Screen



Plant	Tariff Item	Acronym	Unit	From Date	To Date	Year	No. of Years In Open	Value	Edit	Delete
	Capital Recovery Charge Rate	WCRSRy	QAR/1000m3/hour	2020-01-01	2020-12-31	2020	4			
	Contract Capacity	WCCy	1000m3/hour	2020-01-01	2020-12-31	2020	4			
	Fixed Operation and Maintenance Charge Rate	WFOMSRya	QAR/1000m3/hour	2020-01-01	2020-12-31	2020	4			
	Foreign Component of Capital Recovery Charge Rate % WCFy		%	2020-01-01	2020-12-31	2020	4			
	Foreign Component of Fixed Operation and Maintenance WFFy		%	2020-01-01	2020-12-31	2020	4			
	Foreign Component of Variable Operation and Maintenance WVFy		%	2020-01-01	2020-12-31	2020	4			
	Local Component of Capital Recovery Charge Rate % i WCLy		%	2020-01-01	2020-12-31	2020	4			
	Local Component of Fixed Operation and Maintenance WFLy		%	2020-01-01	2020-12-31	2020	4			
	Local Component of Variable Operation and Maintenance WVLy		%	2020-01-01	2020-12-31	2020	4			
	Variable Operations and Maintenance Charge Rate	WVOMSRya	QAR/1000m3	2020-01-01	2020-12-31	2020	4			

Figure 20- Tariff Configuration Screen

Clients

Eltrix Billing system is a web-based solution; operator can access the Billing system client throughout the network using browsers such as Mozilla Firefox / Chrome / Edge, etc. with restricted user access. Client system can be opened using url <https://Server-IP-Address:8080/EltrixPortal>.

4.6 Eltrix Historian

Historian is a real-time data historian for acquiring, storing, and displaying large amounts of operations and engineering information. It collects data from Eltrix-Communicator deep within the enterprise and stores the plant details, configuration details and output results. It enables accurate and timely performance-based decisions that improve operating costs and it holds the stored information for long-term archiving.

Historian stores the parameters relevant for identifying the Tariff Calculated in each month/year. Along with the process parameter it continuously records the status of plant.

4.7 Security

The Security is provided based on User roles for the configuration and operation of the system. There shall be three or more tiers of roles viz., Administrator, Engineer and Operator (If required privilege based new user groups can be added). Each role has predefined rights to access various modules of the software. All users created under each role inherit the rights and privileges from the role. The access of Users is password protected.

The software has additional features that can track changes,

- 1) If any user changes the configuration, then an automatic email is sent to all the Stakeholders notifying such change. This is shown in below figure.

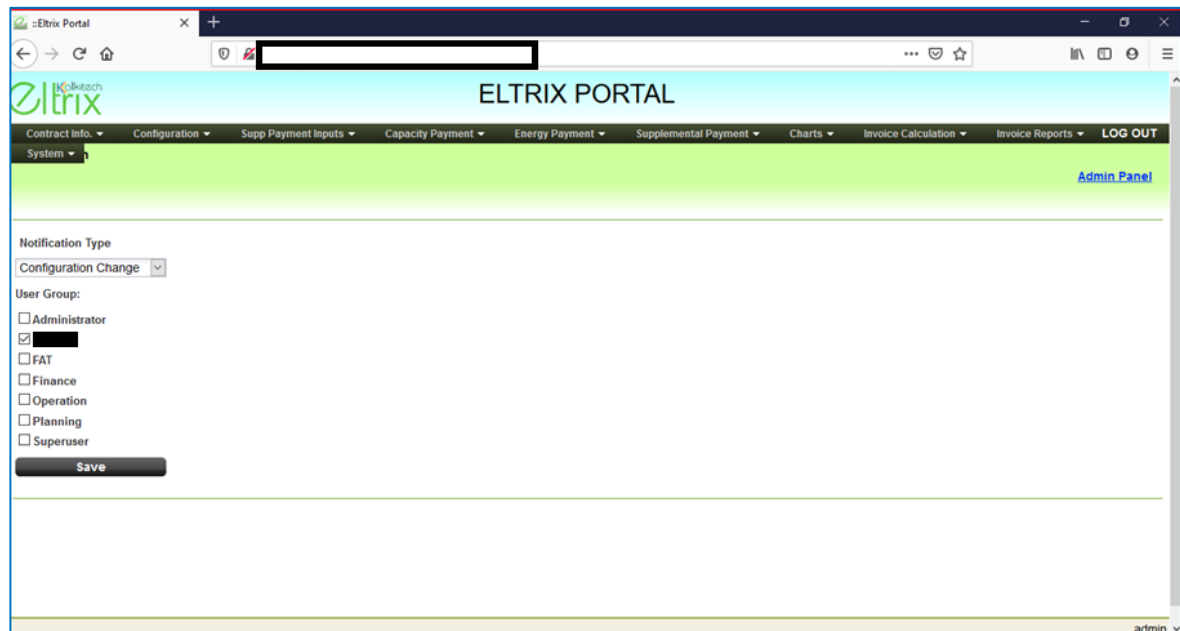
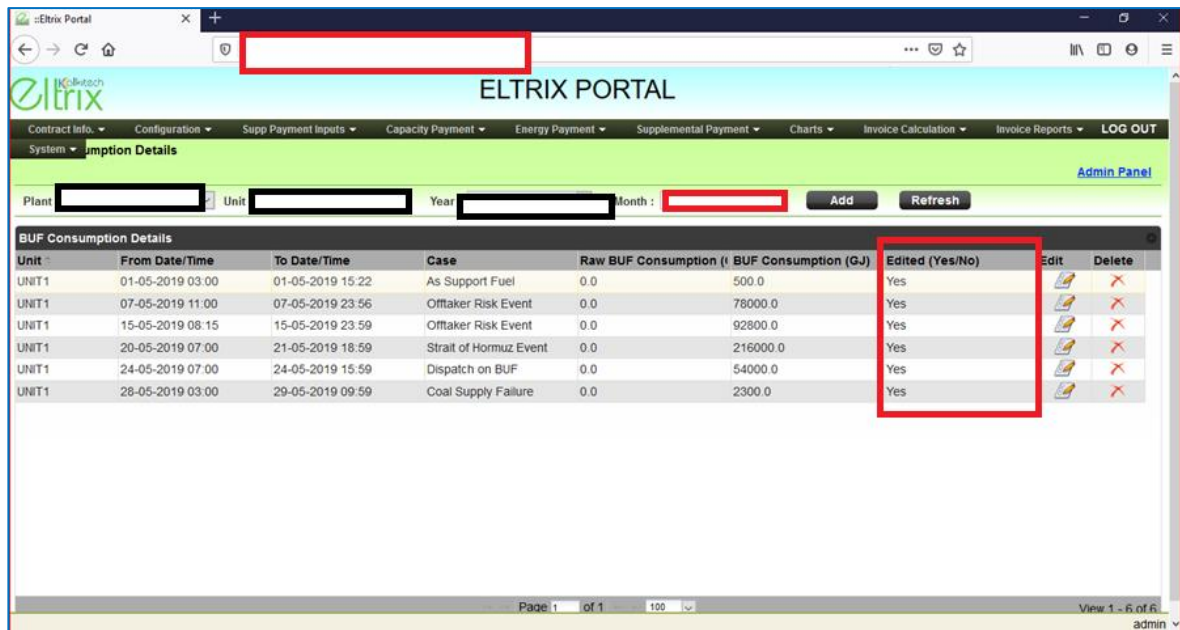


Figure 21- Configuration Change Alert

- 2) If any user edits the value, the same is flagged as shown in below figure.



Unit	From Date/Time	To Date/Time	Case	Raw BUF Consumption (t)	BUF Consumption (GJ)	Edited (Yes/No)	Edit	Delete
UNIT1	01-05-2019 03:00	01-05-2019 15:22	As Support Fuel	0.0	500.0	Yes		
UNIT1	07-05-2019 11:00	07-05-2019 23:56	Offtaker Risk Event	0.0	78000.0	Yes		
UNIT1	15-05-2019 08:15	15-05-2019 23:59	Offtaker Risk Event	0.0	92800.0	Yes		
UNIT1	20-05-2019 07:00	21-05-2019 18:59	Strait of Hormuz Event	0.0	216000.0	Yes		
UNIT1	24-05-2019 07:00	24-05-2019 15:59	Dispatch on BUF	0.0	54000.0	Yes		
UNIT1	28-05-2019 03:00	29-05-2019 09:59	Coal Supply Failure	0.0	2300.0	Yes		

Figure 22- Edit Trail

Remote Support

Eltrix is a web-based application. Kalkitech can provide remote support within an organization security framework.

4.8 Data Validation

The end user can define rules for Validation and Eltrix software provides various tools to implement the same. Based on agreed rules, the missing data is filled in for events like Small Missing data due to device failure, communication failure or due to range and quality check issue. The rule is defined by the customer and engineered during the project implementation. Once approved, the same is locked and changes are permitted only by authorized person. Even if the authorized person makes the change the software can send alerts like email alert to notify the stake holders. All validated data is shown in a separate audit trail.

The following libraries are available for customer to choose from

- Linear Interpolation
- Artificial Neural Network (an AI tool where system gets trained and predicts missing data based on current plant condition)
- Median method (where there is multiple measurement for the same data)

- d) Curve fitting
- e) Any custom calculation can also be implemented.

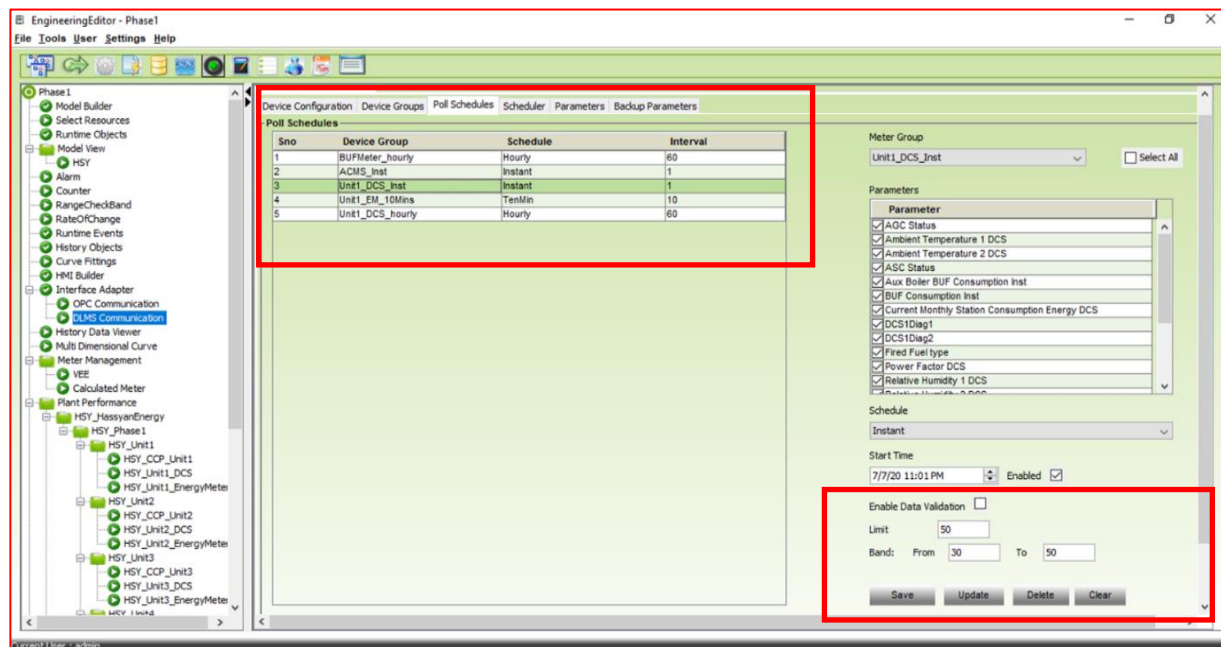
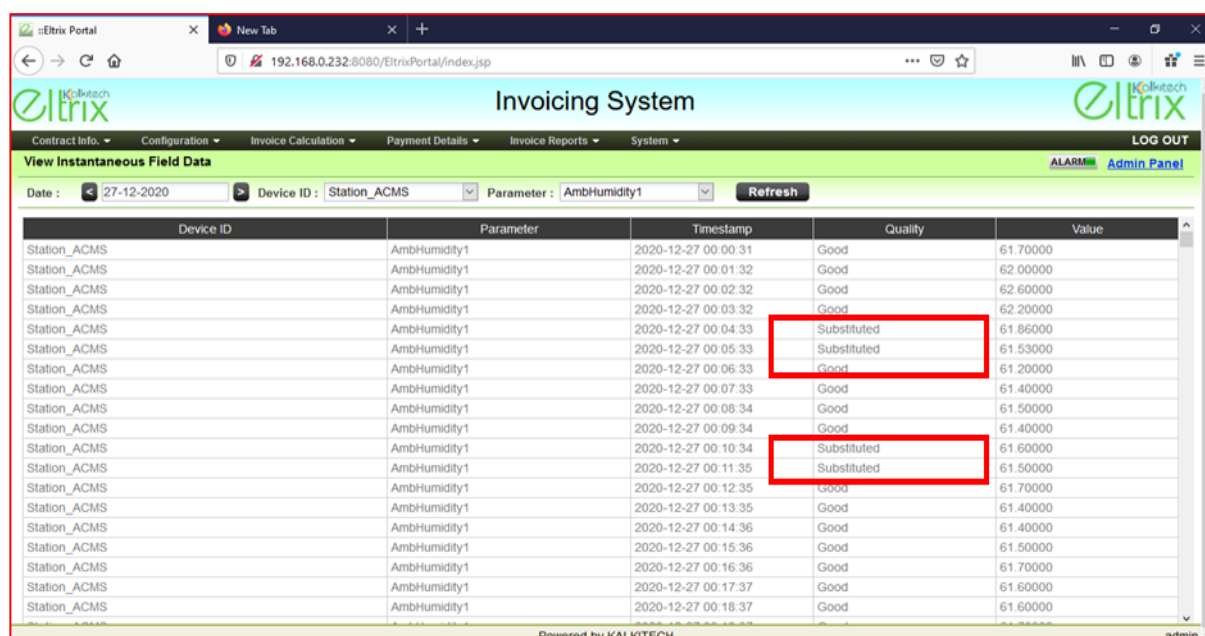


Figure 23- Data Validation Settings



The screenshot shows the 'iEtrix Portal' Invoicing System. The 'View Instantaneous Field Data' section displays a table with columns: Device ID, Parameter, Timestamp, Quality, and Value. The table shows data for Station_ACMS with parameters like AmbHumidity1. Several rows show 'Substituted' values, which are highlighted with red boxes.

Device ID	Parameter	Timestamp	Quality	Value
Station_ACMS	AmbHumidity1	2020-12-27 00:00:31	Good	61.70000
Station_ACMS	AmbHumidity1	2020-12-27 00:01:32	Good	62.00000
Station_ACMS	AmbHumidity1	2020-12-27 00:02:32	Good	62.60000
Station_ACMS	AmbHumidity1	2020-12-27 00:03:32	Good	62.20000
Station_ACMS	AmbHumidity1	2020-12-27 00:04:33	Substituted	61.86000
Station_ACMS	AmbHumidity1	2020-12-27 00:05:33	Substituted	61.53000
Station_ACMS	AmbHumidity1	2020-12-27 00:06:33	Good	61.20000
Station_ACMS	AmbHumidity1	2020-12-27 00:07:33	Good	61.40000
Station_ACMS	AmbHumidity1	2020-12-27 00:08:34	Good	61.50000
Station_ACMS	AmbHumidity1	2020-12-27 00:09:34	Good	61.40000
Station_ACMS	AmbHumidity1	2020-12-27 00:10:34	Substituted	61.60000
Station_ACMS	AmbHumidity1	2020-12-27 00:11:35	Substituted	61.50000
Station_ACMS	AmbHumidity1	2020-12-27 00:12:35	Good	61.70000
Station_ACMS	AmbHumidity1	2020-12-27 00:13:35	Good	61.40000
Station_ACMS	AmbHumidity1	2020-12-27 00:14:36	Good	61.40000
Station_ACMS	AmbHumidity1	2020-12-27 00:15:36	Good	61.50000
Station_ACMS	AmbHumidity1	2020-12-27 00:16:36	Good	61.70000
Station_ACMS	AmbHumidity1	2020-12-27 00:17:37	Good	61.60000
Station_ACMS	AmbHumidity1	2020-12-27 00:18:37	Good	61.60000

Figure 24- Data Substitution Audit Trail

4.9 Data Granularity

Eltrix PASS supports calculation with data granularity of every minute. This helps to understand the plant status near Realtime. The specification asks for data granularity of 6 minutes which can be met.

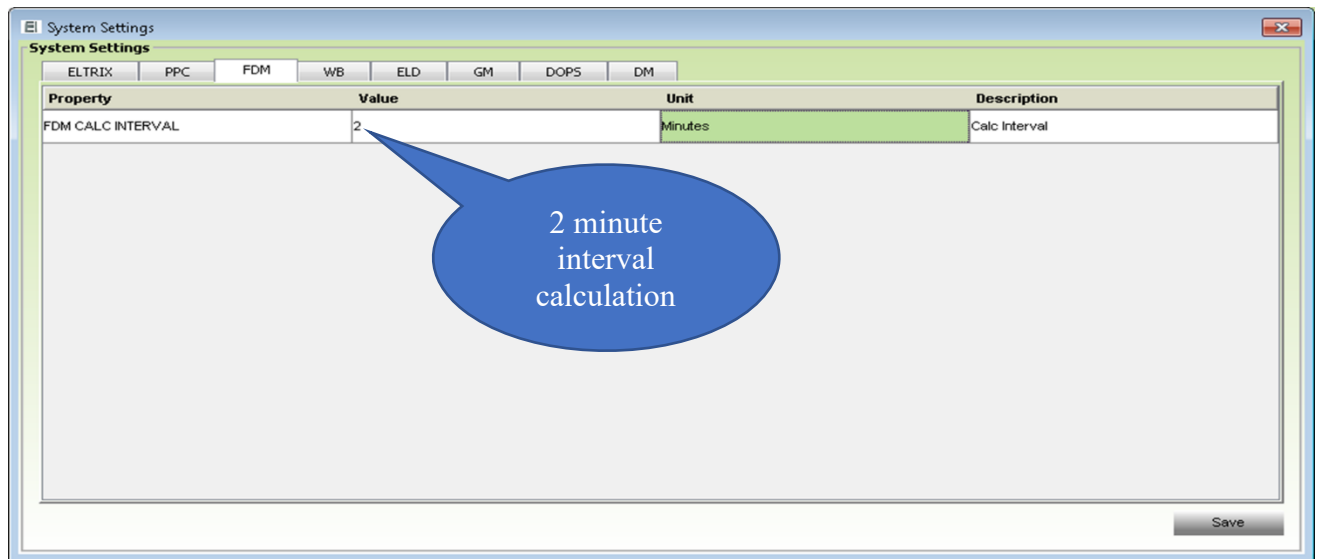


Figure 25- 2 Minute Calculation Interval

4.10 Invoice Calculation (Sample)

The Payment to the Company might consists of,

1. Energy Payments in respect of Net Electrical Energy
2. Deemed Energy Payment in respect of Deemed Net Electrical Energy

$$TP_n = ((NEE_{n,act} + NEE_{n,deemed}) * EC_n / 100)$$

TP_n	=	Total payments for Net Electrical Energy in Billing Period n
NEE_{n,act}	=	Net Electrical Energy as delivered to and metered at the Connection Point in Billing Period n (kWh)
NEE_{n,deemed}	=	Deemed Net Electrical Energy in Billing Period n (kWh)

EC_n	=	Energy Charge to be used for Billing Period n, calculated and adjusted as mentioned in the PPA
N	=	Billing period

Energy Charge

The Energy Charge to be used for billing period n in the calculation of the energy payment is,

$$EC_n = CCR_n + OMR_n$$

EC_n	=	Energy Charge to be used for Billing Period n (Hals/kWh)
CCR_n	=	Capital cost recovery component of Energy Charge in Billing Period n
OMR_n	=	O&M cost recovery component of Energy Charge in Billing Period n
n	=	Billing period

Deemed Net Electrical Energy

The Deemed Net Electrical Energy of the plant in billing period (n) is determined as follows and will be calculated as specified in the PPA.

$$NEE_{n,deemed} = \sum_{i=1}^I NEE_{i,deemed}$$

$NEE_{n,deemed}$	=	Deemed Net Electrical Energy in Billing Period (n) (in kWh)
$NEE_{i,deemed}$	=	Capital cost recovery component of Energy Charge in Billing Period n
i	=	Deemed Net Electrical Energy Interval
I	=	Number of Deemed Net Electrical Energy Intervals in Billing Period (n)

4.11 Audit

During the FAT and/or during the commissioning, customer can engage an independent third party to perform model audit of the PPA calculation. Generally, Kalkitech will prepare its scenarios, similarly the independent auditor and end user can also prepare and test the scenarios to confirm the correctness of the model. FAT is conducted using simulated inputs and SAT can be conducted with live inputs.

5 Hardware and Software Requirements

HW and SW (OS, etc.) will be discussed and finalized in the engineering phase.

5.1 Eltrix Billing System Server Machine

S.No	Item	Hardware Description
1	Hardware (Model)	Dell Power Edge R440 OR Better
2	Form Factor	Rack Mount
3	Power	2x 550 W Hot-Plug with Full Redundancy
4	Processor	Intel Xeon 4208 2.1GHz 8C/16T
5	Memory	16GB RDIMM, 2666MT/s
6	RAID	PERC H730 Raid Controller
7	Hard Disks	2x 600 GB 15K RPM SAS 12 Gbps 2.5” Hot-Plug HDD
8	Drive	16x DVD-ROM
9	Console	Rack Console, Monitor, KBD, Mouse / KVM
10	Operating System	Microsoft Windows Server 2019 Standard
11	Software	Excel 2019
12	Anti-Virus	McAfee OR other
13	Backup Software	Veritas Backup Exec OR Veeam

6 BOM

S. No	Item	Location	Scope	Quantity
PASS System				
1	Eltrix PASS Software License with 10 User Web Clients	• 1x Settlement system at plant facility (Online PSS) • 1x Settlement system at SPPC premises (Offline PSS)	Kalkitech	02 Nos.
2	Settlement System Server	• 1x Server at plant (1x for Online PSS) • 1x Server at SPPC premises (1x for Offline PSS)	Kalkitech	02 Nos.
3	PSS's KVM switch with monitor	• 1x for Online PSS	Kalkitech	01 No.
4	PSS Panel with Accessories	• 1x for Online PSS	Kalkitech	01 No.
5	Keyboard, Mouse & Monitor	• 1x for Offline PASS	Kalkitech	01 No.
Meter Data Acquisition (MDA)				
6	MDA Software License	• 2x Server (Redundant at meters end)	Kalkitech	02 Nos.
7	MDA's database server	• 2x Server (Redundant at meters end)	Kalkitech	02 Nos.
8	MDA's SYNC 3000	• 2x SYNC 3000 (Redundant)	Kalkitech	02 Nos.
9	MDA's KVM switch with monitor		Kalkitech	01 No.
10	MDA Panel with Accessories		Kalkitech	01 No.
11	Network Switch	Industrial Grade	Kalkitech	02 Nos.
12	Networking Cables	As mentioned in the Figure -1 Solution architecture	Kalkitech	As required



Third party software and support				
13	Operating Systems	in Server and WS	Kalkitech	As required
14	MS Excel	in server and WS	Kalkitech	As required
15	Firewall	at plant location	Kalkitech	02 Nos.
16	Warranty	Regular support as per Section 12.	Kalkitech	12 months
17	Annual Maintenance Support after Warranty	Offered separately.	Kalkitech	

Note: BOM shall vary after verification of the requirements of the Client / Offtake

7 Hardware/Software Assumptions & Exclusions

- Supply as per BOM in Section 6.
- Forecasting system and any forecast values for computation are assumed to be client scope.
- It is assumed that the input/data required for design and engineering is provided before the acceptance test.
- It is assumed that the Input tags/measurements what is required to perform Eltrix modules are available at data source and accessible.
- Patch chord cabling within the control room is considered.
- Offline PASS software at SPPC is considered.
- Data Push from DCS for replica server network is assumed to be in client scope.
- The communication between plant to SPPC site and Substation to Plant MDA is assumed to be in client's scope.
- The replica server at SPPC network is assumed to be in client's scope.
- Client shall provide the network and required PSS data from the replica server shall be made available in the offline location at switch from where the data will be fetched by offline PSS and CSS
- The required inputs for Solar Performance Module are assumed to be available in the plant DCS / SCADA.
- All customer expenses for discussions and witness tests are excluded. The expenses include but are not limited to items like travel, boarding lodging, visa, etc.
- The raw data from the DCS to replica server as mentioned in the SPPC Version 7 is assumed to be in client's scope.

8 Project Execution Model

We understand that developing systems in today's evolving technologies is a challenge. We have defined a custom methodology that is suitable to the requirements. This is a well-practiced and time-tested methodology that we have successfully used in many of Kalkitech projects.

The diagram below depicts the project engineering and execution life cycle.

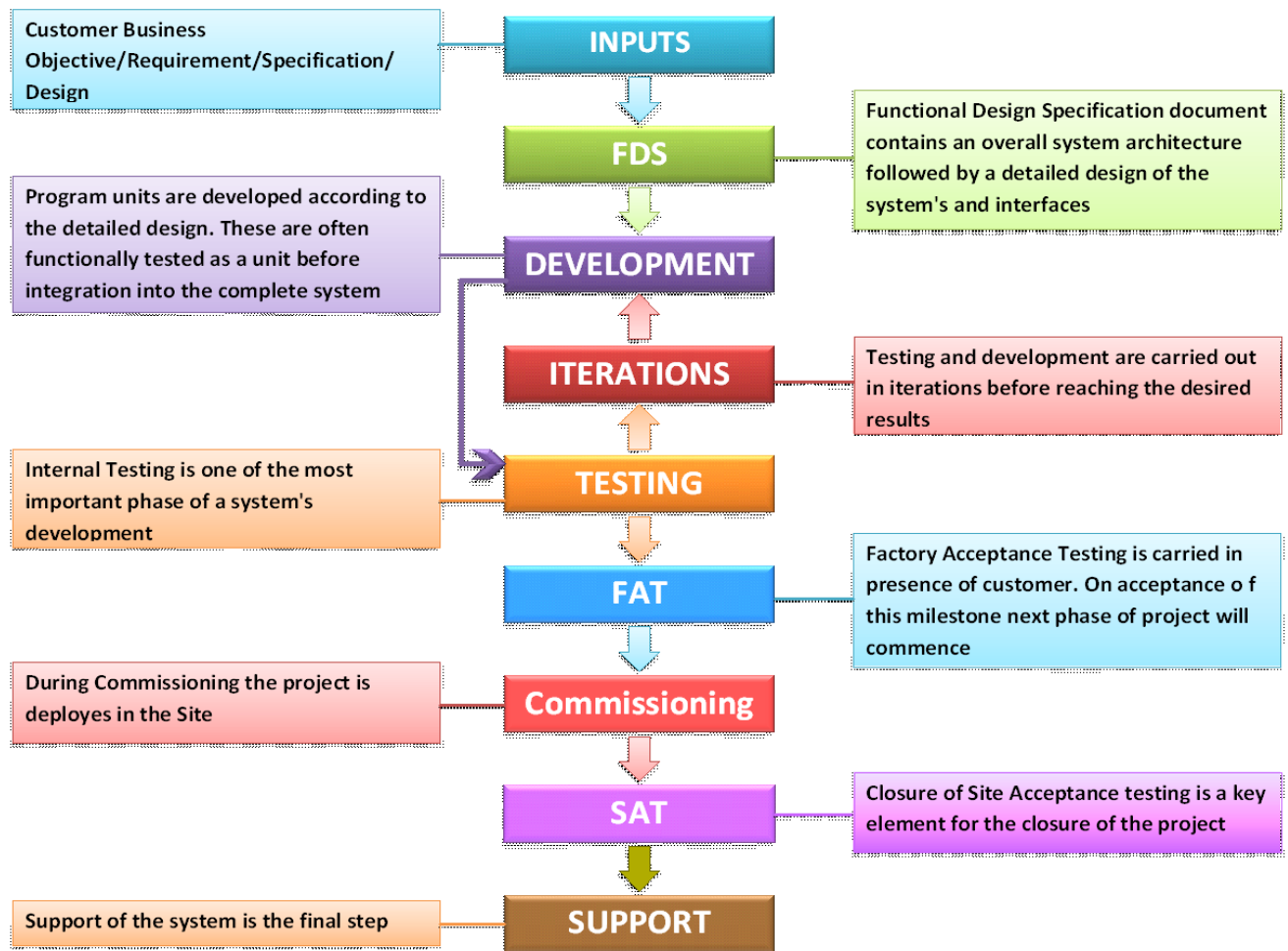


Figure 26- Project Engineering and Execution life cycle

8.1 Highlights of the Model

- Inconsistencies between requirements, design and implementation are detected early and resolved before they manifest as major discrepancies.
- Changes in the business requirements can be accommodated easily due to iterative model thus helping to achieve significant cost benefits.
- Shorter turn-around of systems with the desired functionality
- Dedicated testing team enabling interference-free testing processes for system, integration testing.

Kalkitech will tailor the baseline execution methodology as suited for a particular customer at the beginning of any project. Kalkitech will be flexible to accommodate the processes required to be followed as per customer standards.

9 Project Implementation Plan

The schedule breakdown shall be adjusted finally according to Client requirements.

9.1 Project Stages and Deliverables

Stage	Tasks/Deliverables	Responsible	Support
Study, Planning,	Objective ➤ Come up with a detailed project plan for the project. ➤ Complete review of the requirements provided by customer. Tasks ➤ Specification/Scope Finalization ➤ Input/Data Collection ➤ Study of Inputs ➤ Site Survey, if required ➤ Detailed project plan Preparation ➤ Detailed Design Specification Preparation Deliverable ➤ Project plan ➤ Engineering Design Specification ➤ FAT/SAT Procedure	Kalkitech	Customer
Engineering and Unit Testing	Objective ➤ Plant Modeling ➤ Engineering ➤ Unit Testing ➤ Integration Testing Tasks ➤ Engineering of various modules ➤ Unit level testing of each module	Kalkitech	Customer

Stage	Tasks/Deliverables	Responsible	Support
	➤ Module Integration Testing (System Testing – Internal) ➤ Validation and Verification Deliverable ➤ Tested application package ready for FAT ➤ Internal Acceptance Report		
Factory Acceptance Testing	Objective ➤ Internal System testing ➤ Factory Acceptance Testing ➤ Delivery Tasks ➤ Simulate field Process Inputs and perform end to end testing ➤ Detailed validation and verification based on Approved test Cases - Internal ➤ Prepare for Factory Acceptance Testing ➤ Perform Test as per Approved FAT Procedure ➤ Handover Test Reports and Application Media (Delivery) ➤ User manual Deliverable ➤ Software Media and documentation ➤ Factory Acceptance Document – Mutually Accepted by Customer and KALKI Project Team	Kalkitech	Customer

Stage	Tasks/Deliverables	Responsible	Support
Site Acceptance Testing and Hand over	Objective ➤ Site Commissioning ➤ Testing with Field Values ➤ Acceptance Testing and handover Tasks ➤ Site Installation ➤ Fine tuning System as per site conditions ➤ Detailed validation and verification based on Approved test Cases ➤ Prepare for site acceptance testing ➤ Acceptance testing At Plant as per approved SAT Procedure. ➤ Handover Test Reports ➤ End User Training Deliverable ➤ Demonstration ➤ Acceptance Signoff ➤ Training Documentation	Kalkitech	Customer/End Customer
Warranty	Objective ➤ Warranty support to the accepted engineered software package which provides peace of mind for the end customer. Details mentioned in section 12 Tasks ➤ Analyze and resolve post implementation issues. ➤ Monitor health of the system ➤ Verification of results. Deliverable ➤ Email/Telephonic/Remote Support	Kalkitech	

Stage	Tasks/Deliverables	Responsible	Support
	➤ Bug Fixes		
Annual Maintenance and Support (Optional)	Objective ➤ Continuous support of the application and ensure high value support. Details in section 12 Tasks ➤ Periodic updates ➤ Helpline support ➤ Assured problem resolution Deliverable ➤ Email/Telephonic/Remote Support ➤ Bug Fixes ➤ Patches/Upgrades	Kalkitech	Customer

9.2 Schedule

It requires 6-9 months to complete the project. The schedule can be mutually worked out.

10 Execution Methodology

Eltrix is a secure web-based architecture. Kalkitech has developed tools that can enable to execute the project remotely, if required. Recently, we have executed one project in UAE in a similar manner.

10.1 Project Organization Structure

The key to the success of the proposed solution from **Kalkitech** lies in its proven engagement governance model. The strength of this model lies in the fact that it has been fine-tuned through application to a very large number of client engagements and is stable and proven. Its key strengths are:

- A clear delineation of roles and responsibilities between Client and **Kalkitech**
- Established communication protocols
- Single/defined point of contact for the client

The following diagram represents this model schematically:

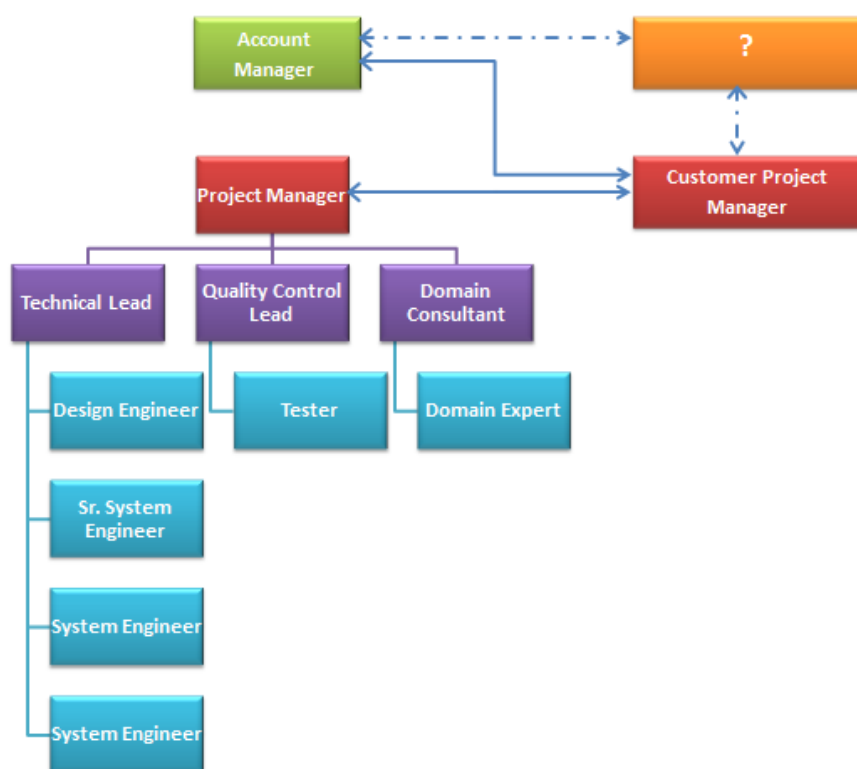


Figure 27- Project Organization Structure

Project or application-level governance focuses primarily on the following aspects:

- Schedule, task, and resource management
- Issue resolution
- Knowledge management
- Quality management
- Monitoring, communications, and reporting

Group / Role-Type	Responsibilities
Kalkitech Project Manager	✓ Responsible for managing customer relationship ✓ Responsible for On time Delivery to Customer requirement ✓ Coordination with customer, Internal Project/Execution Teams and Other Stake Holders ✓ Overall Capacity Planning and Resource Management ✓ Detailed execution planning (including defining key milestones)

Group / Role-Type	Responsibilities
	✓ Resource tracking and work allocation ✓ Risk Management ✓ Status Tracking ✓ Change Management ✓ Customer Primary point of contact, Periodic Status reporting to Customer. ✓ Ensure Knowledge Sharing is done across the organization
Kalkitech Account Manager	✓ Manage Client Relationship at strategic levels ✓ Periodically monitor project related activities and do a status check on the health of the overall engagement ✓ Act as the escalation point for all project related issues which cannot be solved at the project team level
Kalkitech Technical Lead	✓ Resource tracking and work allocation ✓ Engineering and testing ✓ Test the delivery before delivering it to onsite team ✓ Regularly monitor the progress of each request and report status to Project manager ✓ Periodic status reports to help monitor the progress of changes or enhancements efforts ✓ Capture and analyze various metrics ✓ Ensure timely and quality deliveries ✓ Resolve all issues ✓ Facilitate preparation and delivery of all deliverables ✓ Interact with the Quality groups to ensure that the highest level of quality standards are maintained
Kalkitech Project Team Members	✓ Analysis, Design, Development, Testing and Deployment Support ✓ Providing bug fixes and maintenance tasks ✓ Support ✓ Maintaining records of Incident Requests and associated metrics
Kalkitech Quality Leader	✓ Assist the project team in tailoring the quality process as per needs ✓ Help in process improvements and refinements. ✓ Perform regular audits to ensure process compliance by the project team

Group / Role-Type	Responsibilities
	✓ Raise early warnings to project manager, project leader in case of process or delivery issues ✓ Help project team in defect analysis and prevention ✓ Help project team in metrics consolidation and analysis
Kalkitech Domain Group	✓ Provide required Domain support ✓ Help the team with domain knowledge in design and engineering ✓ Provide Technology and Architecture Support

10.2 Project Management

Kalkitech follows a well-defined Project Management Life Cycle, engaging pre-defined specialists at various stages within the Kalkitech partnership that leverages our experience in handling large projects. The project management process has been developed to address all the risks and needs attached to a remote execution and virtual team model, where in teams work from on-site support as required. Kalkitech provides well defined **status reporting** mechanism in place. Besides providing periodic **Status Reports**. These mechanisms have proven to be very effective in projects Kalkitech has undertaken to date.

10.2.1 Project Tracking

Standard Project Management tools will be used for Project Management and Tracking.

Content	PMP Project Health Metrics capture ➤ Schedule Variance ➤ Defect Density for each phase of the Project
----------------	---

10.2.2 Project Status Reporting

We provide status reports at a pre-decided frequency. It is weekly for small duration projects and fortnightly for large projects. The following table summarizes the contents of the Status Report.

Frequency	Weekly
Content	• Work Planned for the last reporting period • Work Accomplished • Work due in the next reporting period • Tasks on-going • Unresolved Issues • Major Risks
Mode of Communication	• E-Mail • Telephone

10.3 Quality Management

Quality is an intrinsic component of Kalkitech's project execution methodologies. Kalkitech services are based on a tried, tested and trusted Quality Management System that has evolved over years and has earned international certifications like ISO 9001. The Kalkitech quality assurance team focuses on enhancement of the various processes and standards of the quality system.

10.3.1 Quality Assurance Process – An Overview

An overview of the existing Quality Assurance processes is detailed below. Projects are bound within the framework of the company's Quality Management System (QMS). Kalkitech's QMS requires compliance with the following:

- Project Management Plan (PMP) – Planning & Schedules, Quality Plan and Configuration Management
- Requirements Specifications Document
- Detailed Design Document
- System Test Plan

The Quality of the Work Product is controlled by:

- Review of all Technical Specifications Documents
- Review of Engineering
- Review of Test Plans and Procedures
- Review of Testing

The company's Central Quality Assurance (CQA) Department conducts frequent Audits and Reviews to ensure compliance to established processes and their revision as needed.



Kalkitech has a mature Metrics Program. Under this program productivity and defect data is collected from all projects and analyzed and then feedbacks and actions are taken as needed.

These are consolidated and analyzed for:

- Better Monitoring & Control of Projects with respect to Efforts, Schedules and Quality
- Validating and Refining the Kalkitech Estimation Processes
- Monitoring and Setting Improvement Baselines for Productivity and Defect Densities

Kalkitech User Acceptance Guarantee

During this phase Kalkitech will assist customer in bug resolution. All defects which are documented in bug tracking tool will be analyzed and categorized into the following categories.

- Critical - No functional workaround. Causes testing to stop (“Critical”)
- Serious - Very painful functional work around. Only acceptable as a temporary fix during testing (“Serious”)
- Moderate - Functional work around. Something that the user will be able to live with for a limited time (“Moderate”).
- Cosmetic - No impact on functionality and the functional user will never see it (“Cosmetic”).

At the end of User Acceptance Test, Kalkitech will ensure that:

There are no Severe or Critical bugs, as defined above, 95% of reported bugs are resolved. Less than 35% of open bugs are Moderate, as defined above. All the open bugs will be worked during the warranty time period.

10.4 Change Management

For successful execution of any project, an effective Change Management process needs to be in place.

10.4.1 Change Management Process

The following flowchart illustrates the Change Management process that **Kalkitech** uses to manage change efficiently.

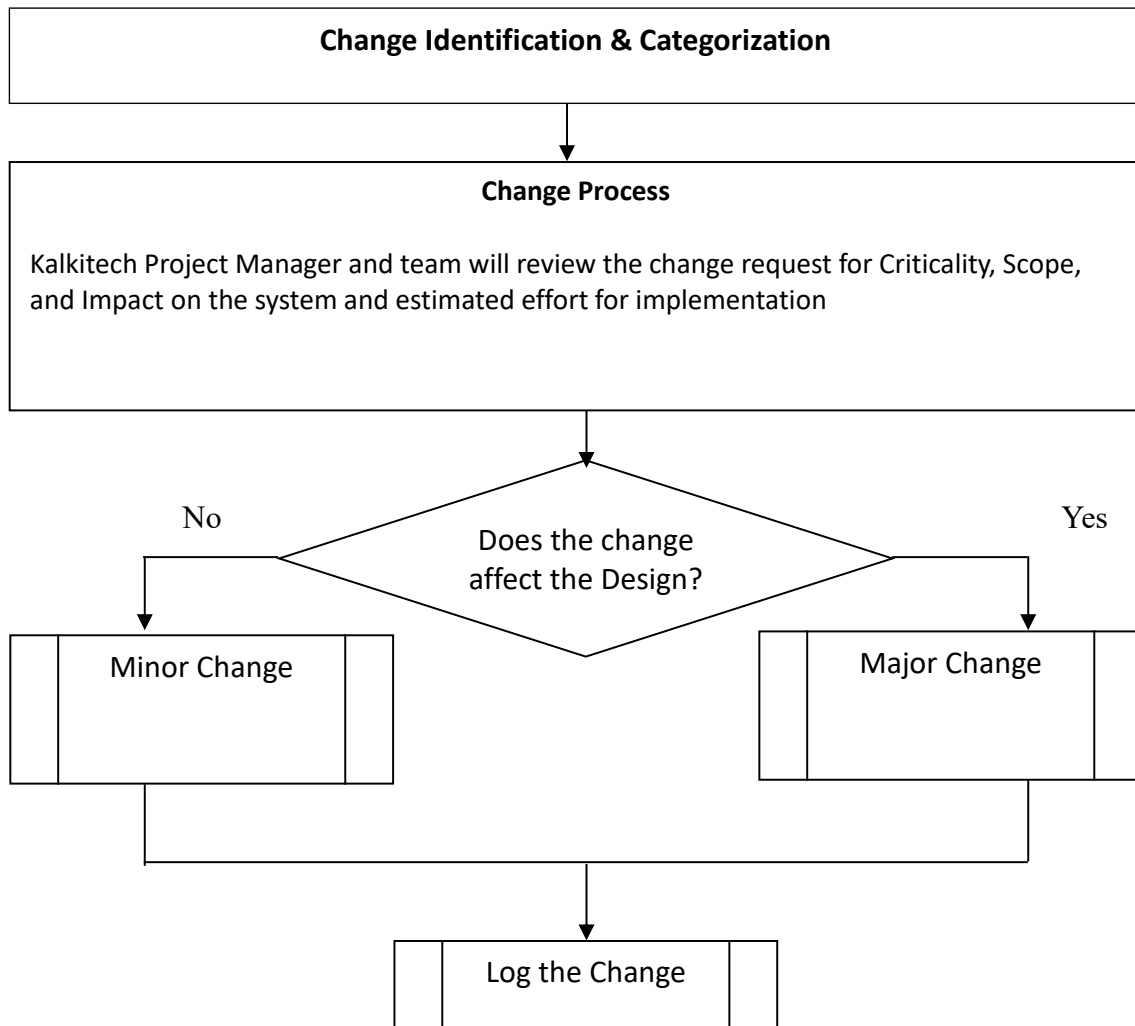


Figure 28- Change Management

Change Identification and Categorization

Changes requested by the customer will be identified and categorized into

- Additional features
- Changes in specification of existing features
- Changes in approved assumptions, which will impact effort and span
- Changes in version of the software used

10.4.2 Minor Changes

Changes that do not affect the design will be classified as **minor**. Our iterative engineering methodology absorbs these changes as a normal part of engineering/development.

10.4.3 Major Changes

Changes that affect the design will be classified as major. We will provide the Impact Analysis and effort/cost estimate reports to **Customer** for such changes. Based on these inputs, **the Customer** Project Manager will take the decision to include/exclude the change.

10.4.4 Change Log

A Change Log will be maintained to log the change request, date of change, priority etc. An entry will be made into the Change Log irrespective of the acceptance or non-acceptance of the change.

10.5 Risk Management

Risk identification, evaluation and mitigation plan is very critical in the project monitoring and control. We as an organization takes special attention in identifying the risk at early stage, analyzing it and prepare for it so that it should not give any surprise during execution and there is already a plan put in place to mitigate the risk.

Kalkitech categorizes potential risks into 4 types:

10.5.1 Schedule Risk:

Any risk identified which has an impact on the schedule is captured under the above category.

Below are the major risks identified under the above category:

- Delay in Input from Customer.

- Delay in Document Approvals.
- Delay in Internal/External Product Delivery
- Delay in Testing and Bug Fixes
- Delay in Site/Customer Readiness
- Delay/Issues in Travel – VISA, Work Permit (Long term Site Activities) etc.

10.5.2 Resource Risk:

Potential Risks identified in connection with resources who are working for the project are captured.

Below are the major risks identified under the above category:

- Availability of trained/skilled resource for the project.
- Shared Resources – Multiple activities impacts the overall plan.

10.5.3 Commercial Risk:

Potential Risks identified which has an impact on the project commercials.

Below are the major risks identified under the above category:

- Delay in billing confirmation from customer
- Delay in Payments
- Unplanned travel
- Travel Confirmation without advance notice
- Payment Terms – Retention payment for site/Advance Payment to vendors
- Financial Charges – BG etc.

10.5.4 Technical Risk

Potential Risks identified which has an impact on the solution are captured. Risks listed under this can cause delay in delivery which will in turn result in commercial risk, i.e., overall project risk is envisaged if not identified and planned properly.

Below are the major risks identified under the above category:

- Interface module issues
- Data Source connectivity issues



- Hardware issues
- Lack of clarity in scope
- Lack of availability of inputs required to meet customer expectations.

Kalkitech will identify risk with respect to the project, evaluate and prepare mitigation plan as part of the Project Plan.

11 Training and Knowledge Transfer

A detailed training is planned as part of the project execution. Training schedules will be planned during the acceptance stages of the project.

12 Warranty Services

Kalkitech warrants that, for a period of 12 Months from the user acceptance, the supplied software will perform in substantial conformance with the applicable software documentation. The warranty is subject to exceptions for software problems that are not caused by software defects or are otherwise beyond Kalkitech's control.

During the Warranty period,

- **Customer** shall notify the Kalkitech designated support personnel/corporate support via email of a defect. The email shall contain a description, severity level, functionality affected, and steps and data needed to reproduce the defect.

Kalkitech offers 2 types of warranty for duration of 12 months from the date of user acceptance, Software warranty which comes with the base software package called **Regular Warranty Support** and Optional Software warranty which comes with enhanced services, at an additional cost, called **Premium Warranty Support**. Customer can opt for either of the above based on the business needs. Support will be extended during the business hours of Kalkitech.

Business hours of Kalkitech is defined as:

- Monday – Friday: 09:00 AM to 06:00 PM
- Support will be restricted to working days of the organization excluding, all public holidays/holidays defined by Kalkitech



12.1 Types of Warranty Service:

Regular Support:

During the regular warranty period,

- Kalkitech shall respond to the complaint/query within 48 business hours of notification.
- Kalkitech shall correct bugs and document the details of the issue and solution/suggestions, ensuring that the software is working in accordance with the agreed functionalities at the time of user acceptance and applicable documentations. Mode of support will be via email/telephone/remote sessions.

Premium Support:

Customer can avail premium support, at an additional cost. KALKI will charge customer, **25%** of the purchase order value, during the contract period to avail the premium warranty support.

Following services will be covered:

1. Kalkitech shall respond to the complaint/query within 24 business hours of notification,
 2. Solutions/Suggestions given for the defects reported will be recorded and shared with customer for historic analysis.
 3. In Addition to the remote support on need basis, periodic Site visit to conduct the complete audit of the Software deployed (Software Health Checkup) to ensure that the software is working as expected - Periodicity will be defined case to case.
 - 1 Day per quarter per location – limited to 3 days at site
 - All travel/expenses will be to the account of customer, unless covered in the contract
 4. System Cleanup – Will be done during the proposed health checkup visit.
 5. Defect resolution
 6. Calculation Revisions – Due to change in inputs
 7. Software Re-Installation Support due to hardware changes.
- Any additional man days spent at site/additional travel beyond the inclusions mentioned above, will be charged extra.

Following services **will not** be covered under the scope of Warranty Support described above:



2. Changes resulting due to the operational scenarios.
3. Changes resulting due to infrastructure limitations such as bandwidth, disks running out of space etc.
4. Changes due to hardware failure/replacement
5. Data Loss due to the malfunctioning of the data source.
6. Deletion of Application files/Alteration of existing configuration.
7. Errors Caused due to installation of any third party software.
8. Errors due to interaction with other software.

13 Maintenance - Post warranty

Kalkitech will provide **Customer** the support and maintenance services for the sole purpose of assisting and supporting Customer in using the Software in accordance with this Agreement. Customer can opt for either of the service proposals mentioned in the warranty section and continue availing the services from the date of completion of warranty support for a mutually agreed duration at an additional cost.

Customer Responsibilities during maintenance period:

With respect to the support and maintenance services described above, Customer shall:

- Designate support contact persons who can be reasonably expected to work with Kalkitech in the resolution of problems. A primary technical support person must be named along with a back-up.
- Actively assist in problem identification, investigation and resolution.
- Make reasonable efforts to resolve problems prior to contacting Kalkitech.
- Keep records of support and problem resolution, and use these records in determining problem resolutions and/or workarounds.
- Comply with problem reporting procedures that may be reasonably set by Kalkitech from time to time.

Kalkitech Responsibilities.

With respect to these support and maintenance services described above Kalkitech shall:

Provide access to designated employees of customer to qualified Kalkitech support personnel during normal business hours via telephone or other means of electronic communications for the purpose of answering questions and resolving problems regarding the use of the Software by **Customer**.

- Use reasonable efforts in accordance with the severity of the problem to investigate, correct or assist in the correction of reported problems and assist in the development of a work-around.
- Maintain records of reported problems, associated causes, fixes, or corrections

and to make such information available to customer.

- Ensure that the fix applied to address the reported issue is not impacting any of the existing functionality/performance of the delivered system.

- Turnaround time for issues/queries and the scope of work included in the warranty is as per the agreed option during the contract finalization from the below mentioned.

Regular/Standard Maintenance/Subscription

Services covered under Regular/Standard Maintenance are same as the regular warranty, stated above. KALKI will charge customer, **18 %** of the purchase order value per annum to avail this service.

- Kalkitech shall respond to the complaint/query within 48 business hours of notification.
- Kalkitech shall correct bugs and document the details of the issue and solution/suggestions, ensuring that the software is working in accordance with the agreed functionalities at the time of user acceptance and applicable documentations. Mode of support will be via email/phone/remote sessions – depending on the criticality of the issue.

Premium Maintenance/Subscription

Customer can avail premium support, at an additional cost. KALKI will charge customer, **25%** of the purchase order value per annum to avail the premium maintenance service.

Following services will be covered:

- Kalkitech shall respond to the complaint/query within 24 business hours of notification,
- Solutions/Suggestions given for the defects reported will be recorded and shared with customer for historic analysis.
- In Addition to the remote support on need basis, periodic Site visit to conduct the complete audit of the Software deployed (Software Health Checkup) to ensure that the software is working as expected - Periodicity will be defined case to case.
 - 1 Day per quarter per location, for a duration of 1-3 days at site
 - All travel/expenses will be to the account of customer, unless covered in the contract.
- System Cleanup – Will be done during the proposed health checkup Visit.



- Defect resolution
- Calculation Revisions – Due to change in inputs
- Software Re-Installation Support due to hardware changes.

Any additional man days spent at site/additional travel beyond the inclusions mentioned above, will be charged extra.

Following services **will not** be covered under the scope of Maintenance Support described above:

- Enhancements due to changed requirements.
- Changes resulting due to the operational scenarios.
- Changes resulting due to infrastructure limitations such as bandwidth, disks running out of space etc.
- Changes due to hardware failure/replacement
- Data Loss due to the malfunctioning of the data source.
- Deletion of Application files/Alteration of existing configuration.
- Errors Caused due to installation of any third-party software.
- Errors Caused due to installation of any third-party software.



14 Eltrix references for Billing and Settlement Systems.

(S. No. in below table corresponds to the S. No. of complete Eltrix references in Section 15)

S.No	End Customer	Plant Name	Number of Units	Name of OEM/Customer	DAS/DCS System	Solution Provided	Year of Implementation	Status
1	Shuaibah 3	Shuaibah 3 IWP	600,000 m ³ /d	Schneider's Order to Arabian Network Support Co.	-NA-	Plant Settlement System	Dec-23	Design Stage
2	Yanbu 4	Yanbu -4 IWP	450,000 m ³ /d	Engie's Order to Arabian Network Support Co.	-NA-	Plant Settlement System	Oct-23	FAT Completed
3	Jubail 3B	Jubail-3B IWP	570,000 m ³ /d	Engie's Order to Arabian Network Support Co.	-NA-	Plant Settlement System	Nov-23	FAT Completed
4	Ar-Rass PV IPP	Ar-Rass IPP	700 MW Solar PV project	L&T's Order to Arabian Network Support Co.	-NA-	- Plant Settlement System - SPM	Sep-23	FAT Completed
5	Marafiq Red Sea Project	Marafiq Red Sea Project	Multiple Plants	ACWA's order to Arabian Network Support Co.	-NA-	Central Settlement System	Aug-23	FAT Completed
6	SWPC (Saudi Water Partnership Company)	Central Settlement System for all Plants	All Water and Sewage Plants	SWPC order to Arabian Networks Support Co.	-NA-	Central Settlement System	Nov-22	Commissioned
7	Taif (SWPC – Cobra)	Taif ISTP	100,000 m ³ /d (Expandable to 270,000 m ³ /d)	Taif IWPC order to Arabian Networks Support Co.	-NA-	Plant Settlement System	Dec-22	Commissioned
8	Jeddah	Jeddah	300, 000 m ³ /d	Jeddah Airport 2	-NA-	Plant Settlement System	Feb-23	SAT Completed

	Airport 2	Airport 2 ISTP	(Expandable to 500,000 m ³ /d)	ISTP order to Arabian Networks Support Co.				
9	Taqnia Energy	Lyla 40 MW Solar, Saudi Arabia	40 MW Solar PV Plant	Taqnia Energy order to Arabian Networks Support Co.	-NA-	Plant Settlement System	Oct-22	Commissioned
10	Noor Energy 1	Noor Energy 1 CSP	950 MW Hybrid Solar Power Plant Phase 1	Noor Energy 1	ABB DCS	Settlement System	July-21	Commissioned
11	Hassyan Clean Coal Power Station Settlement System	Hassyan	4 X 600 MW	Harbin Electric	ABB DCS	Settlement System	Nov-20	Commissioned
12	QEWCA1 and B2 Billing and Settlement System	QEWCA1 and B2	A1: 70 MILD of Potable Water B2: 567 MW and water production capacity of 29.9 MIGD.	QEWCA, Qatar	Ovation	Billing and Settlement System	Jan-23	Handed Over
13	QEWCA	Doha	A2, A3, 36 MIGD Combined Cycle Cogeneration Plant (GT+HRSG+Desalination+Auxiliary Boiler)	YMA, Bahrain	Centum	- Fuel Demand Modeling, Billing and Settlement - Desalination, Plant performance - Combine cycle power plant performance	Sep-15	Commissioned
14	AES	Mong Duong, Vietnam	2 x 521 MW	YKO, South Korea	Centum	- Performance Monitoring System, Billing and Settlement	Nov-14	Commissioned

YEA, Singapore = Yokogawa Singapore

YKO, South Korea = Yokogawa South Korea

YMA, Bahrain = Yokogawa Bahrain

15 Complete Eltrix References

S.No	End Customer	Plant Name	Number of Units	Name of OEM/Customer	DAS/DCS System	Solution Provided	Year of Implementation	Status
1	Shuaibah 3	Shuaibah 3 IWP	600,000 m ³ /d	Schneider's Order to Arabian Network Support Co.	-NA-	Plant Settlement System	Dec-23	Design Stage
2	Yanbu 4	Yanbu -4 IWP	450,000 m ³ /d	Engie's Order to Arabian Network Support Co.	-NA-	Plant Settlement System	Oct-23	FAT Completed
3	Jubail 3B	Jubail-3B IWP	570,000 m ³ /d	Engie's Order to Arabian Network Support Co.	-NA-	Plant Settlement System	Nov-23	FAT Completed
4	Ar-Rass PV IPP	Ar-Rass IPP	700 MW Solar PV project	L&T's Order to Arabian Network Support Co.	-NA-	- Plant Settlement System - SPM	Sep-23	FAT Completed
5	Marafiq Red Sea Project	Marafiq Red Sea Project	Multiple Plants	ACWA's order to Arabian Network Support Co.	-NA-	Central Settlement System	Aug-23	FAT Completed
6	JSW Bhushan Power & Steels	JSW CPP	506 MW power plant	JSW Bhushan Power & Steel	-NA-	Performance Management System	Aug-23	Design Stage
7	NTPC	Khurja	2 X 660 MW	Emerson	Ovation	- Performance Analysis Diagnostics and Optimization (PADO)	Mar-22	Design Stage
8	NTPC	Buxer	2 X 660 MW	YIL, India	Centum	- Performance Analysis Diagnostics and Optimization (PADO)	Jul- 22	Design Stage
9	SWPC (Saudi	Central	All Water and Sewage Plants	SWPC order to	-NA-	Central Settlement System	Nov-22	Commissioned

	Water Partnership Company)	Settlement System for all Plants		Arabian Networks Support Co.				
10	Taif IWPC	Taif ISTP	100,000 m ³ /d (Expandable to 270,000 m ³ /d)	Taif IWPC order to Arabian Networks Support Co.		Plant Settlement System	Dec-22	Commissioned
11	Jeddah Airport 2	Jeddah Airport 2 ISTP	300, 000 m ³ /d (Expandable to 500,000 m ³ /d)	Jeddah Airport 2 ISTP order to Arabian Networks Support Co.	-NA-	Plant Settlement System	Dec-22	Commissioned
12	Al-Khobar Seawater RO 2 Desalination plant	Al-Khobar	600,000 m ³ per day / RO based	YMA	Centum	Plant Management System	Dec-21	Design Stage
13	Taqnia Energy	Lyla 40 MW Solar, Saudi Arabia	40 MW Solar PV Plant	Taqnia Energy order to Arabian Networks Support Co.		Plant Settlement System	Mar-21	Commissioned
14	Noor Energy 1	Noor Energy 1 CSP	950 MW Hybrid Solar Power Plant Phase 1	Noor Energy 1	ABB DCS	Settlement System	July-21	Commissioned
15	Hassyan Clean Coal Power Station Settlement System	Hassyan	4 X 600 MW	Harbin Electric	ABB DCS	Settlement System	Nov-20	Commissioned
16	Cairo West	Cairo West	1 X 650 MW	Honeywell	Honeywell DCS	Performance Monitoring and Optimization	Dec -- 18	Commissioned
17	Assuit PMS	Assuit, Egypt	1 X 650 MW	YMA, Bahrain	Centum	Performance Monitoring	Dec -- 18	Commissioned

						and Optimization		
18	Bibiyana South PMS	Bibiyana South	1 X 400 MW	Schneider, Singapore	SE DCS	Performance Monitoring System	May-18	Commissioned
19	QEWCA1 and B2 Billing and Settlement System	QEWCA1 and B2	A1: 70 MLD of Potable Water B2: 567 MW and water production capacity of 29.9 MIGD.	QEWCA, Qatar	Ovation	Billing and Settlement System	Jan-23	Handed over
20	Nirma PMS	Nirma Cements	1 X 82 MW	ABB, Pune	800 XA	Performance Monitoring System	Feb-18	Commissioned
21	R2IPP - Rabigh PMO	Rabigh	2060 MW -CCPP	YKO, South Korea	Centum	Performance Monitoring and Optimization	Nov-17	Commissioned
22	Song Hau PADO	Song Hau	2 X 600 MW	YEA, Singapore	Centum	Performance Monitoring, Diagnostics and Optimization	Feb-17	Commissioned
23	SM 200 Phase 2 PIS	SEC 210	1 X 150 MW	YKO, South Korea	Centum	Performance Monitoring and Stress Analysis	Mar-17	Commissioned
24	Masinloc PMS	Masinloc	2 X 304 MW	YKO, South Korea	Centum	Performance Monitoring System	May-17	Commissioned
25	Saudi Electricity Corporation	Riyadh PP13 & 14	2 X 825 MW	YMA, Bahrain	Centum VP	Performance Monitoring and Optimization	Oct-20	Commissioned
26	SEWA PMS	SEWA	2 X 75 MW (Oil and Gas Fired plant)	YMA, Bahrain	Centum	Performance Monitoring System	Feb-17	Commissioned
27	EGAT	Mae Moh	660 MW Thermal Power Plant	GE, Malaysia	ALSPA DCS	Performance Analysis Diagnostics and Optimization (PADO)	Oct-16	Commissioned
28	GMR	Raipur	2 x 685 MW (Coal Fired Super Critical)	YKO, South Korea	Centum VP	Performance Analysis Diagnostics and Optimization (PADO)	Jun-16	Commissioned

29	JayPee	Nigrie	2 x 660 MW	YIL, India	-do-	- Performance Analysis Diagnostics and Optimization (PADO) - Merit Order dispatch	May-15	Commissioned
30	NTPC	Sipat - Stage 2	2 x 500 MW	ABB	PGP	- Performance Analysis and Monitoring System - Economic Load Optimization - Root Cause Diagnostic System - LVS for Alarm Annunciation	Feb-11	Commissioned
31	NTPC	Sipat Stage 1 - Training Simulator Package	3 X 660 MW	YIL, India	Centum CS	- Performance Analysis and Monitoring System - Economic Load Optimization - Root Cause Diagnostic System - LVS for Alarm Annunciation	December 2013-	Commissioned
32	NTPC	Kahalgaon - Stage 2	3 x 500 MW	ABB	PGP	- Performance Analysis and Monitoring System - Economic Load Optimization - Root Cause Diagnostic System - LVS for Alarm Annunciation	Dec-10	Commissioned
33	NTPC	Aravali - Stage 1	3 x 500 MW	YIL, India	Centum CS	- Performance Analysis and Monitoring System	Dec 2014-	Commissioned

						- Economic Load Optimization - Root Cause Diagnostic System - LVS for Alarm Annunciation		
34	NTPC	Rihand - Stage 1	2 x 500 MW	YIL, India	Centum CS	- Performance Analysis and Monitoring System - Economic Load Optimization - Root Cause Diagnostic System - LVS for Alarm Annunciation	Nov-10	Commissioned
35	NTPC	Vindhyachal- Stage 1	6 x 210 MW	ABB	PGP	- Performance Analysis and Monitoring System - Economic Load Optimization - Root Cause Diagnostic System - LVS for Alarm Annunciation	Apr-08	Commissioned
36	NTPC	Singrauli - Stage 2	2 x 200 MW	ABB	PGP	- Performance Analysis and Monitoring System - Economic Load Optimization - Root Cause Diagnostic System - LVS for Alarm Annunciation	Apr-08	Commissioned

37	NTPC	Farakka - Stage 1	3 x 200 MW	ABB	PGP	- Performance Analysis and Monitoring System - Economic Load Optimization - Root Cause Diagnostic System - LVS for Alarm Annunciation	May-11	Commissioned
38	NTPC	Unchahar - Stage 1	2 x 210 MW	YIL, India	Centum CS	- Performance Analysis and Monitoring System - Economic Load Optimization - Root Cause Diagnostic System - LVS for Alarm Annunciation	Feb-10	Commissioned
39	NTPC	Korba - Stage 1	3 x 200 MW	YIL, India	Centum CS	- Performance Analysis and Monitoring System - Economic Load Optimization - Root Cause Diagnostic System - LVS for Alarm Annunciation	Nov-09	Commissioned
40	NTPC	Thalcher	2 x 110 MW	YIL, India	Centum CS	- Performance Analysis and Monitoring System - Economic Load Optimization - Root Cause Diagnostic System	Apr-12	Commissioned

						- LVS for Alarm Annunciation		
41	NTPC	Kahalgaon	4X210 MW Thermal Plant	Siemens India	Spectrum	Merit Order Dispatch	Jun-16	Commissioned
42	NTPC	Singrauli - Stage 1	2 x 200 MW	ABB,India	PGP	Merit Order dispatch	Dec-13	Commissioned
43	KOSEP	Yeosu, South Korea	1 x 354 MW(Fluidized Bed Combustion)	YKO, South Korea	Centum	Performance, Analysis, Optimization	Aug-16	Commissioned
44	SPE	Ain Arnat,Algeria	1015.1(1 GT + 1 HRSG + 1 ST)	YKO, South Korea	Centum	Performance, Analysis, Optimization	Oct-18	Commissioned
45	Marafiq	Yanbu	2 x 350 MW (Oil fired thermal power plant)	YMA, Bahrain	Centum	- Performance, Analysis, Optimization - Thermal Audit and Stress Analysis - Class I calculations	Apr-14	Commissioned
46	SEC, SM200-Phase 1	Philippines	1 x 110 MW (FBC)	YKO, South Korea	Centum	- Performance, Analysis, Optimization - Boiler Life time monitoring	May-15	Commissioned
47	SPE	Biskra,Algeria	1 x 456 MW (Gas Turbine)	YKO, South Korea	Centum	-Performance, Analysis, Optimization	Nov-14	Commissioned
48	QEWG	Doha	A2, A3, 36 MIGD Combined Cycle Cogeneration Plant (GT+HRSG+Desalination+Auxiliary Boiler)	YMA, Bahrain	Centum	- Fuel Demand Modeling, Billing and Settlement - Desalination Plant performance - Combine cycle power plant performance	Sep-15	Commissioned
49	UTE	Punte Del Tigre, Uruguay	1 x 500 MW(2+2+1)	YKO, South Korea	Centum	Performance Monitoring System	Aug-15	Commissioned
50	AES	Mong	2 x 521 MW	YKO, South Korea	Centum	Performance Monitoring	Nov-14	Commissioned

		Duong, Vietnam				System, Billing and Settlement		
51	MDEPC	Bahna, Egypt	750 MW Combined Cycle	YMA, Bahrain	Centum	Performance Monitoring System	Dec-14	Commissioned
52	Cairo Electricity Production Company	Giza North, Egypt	1500 MW Combined Cycle	YMA, Bahrain	Centum	Performance, Analysis, Optimization	Jul-16	Commissioned
53	Saudi Electricity Corporation	Shoiba	1200 MW (2 Block (5 GT + 5 HRSG + 1 ST)	YMA, Bahrain	Centum	Performance Monitoring System	Oct-15	Commissioned
54	Saudi Electricity Corporation	Riyadh PP12	1800 MW (2 Block (4 GT + 4HRSG + 1 ST)	YMA, Bahrain	-do-	Performance Monitoring System	Oct-15	Commissioned
55	PGESCO, Egypt	Nubaria	750 MW	YMA, Bahrain	Centum CS	Performance Monitoring System	Sep-12	Commissioned
56	PGESCO, Egypt	El Atf	750 MW	YMA, Bahrain	Centum CS	Performance Monitoring System	Sep-12	Commissioned
57	PGESCO, Egypt	Cairo West	2 x 200 MW (oil based)	YMA, Bahrain	Centum CS	Performance Monitoring System	Sep-12	Commissioned
58	PGESCO, Egypt	Sidi Krir	750 MW	YMA, Bahrain	Centum CS	Performance Monitoring System	Oct-12	Commissioned
59	Reliance Energy	Dahanu	2 x 250 MW	ABB	PGP	Performance Monitoring System	Aug-07	Commissioned
60	TNEB	Mettur	1 x 210 MW	ABB	PGP	Performance Monitoring System	Dec-10	Commissioned
61	TNEB	Tuticorin	1 x 210 MW	ABB	PGP	Performance Monitoring System	Aug-10	Commissioned
62	PSEB	Bhatinda	2 x 210 MW	ABB	PGP	Performance Monitoring system	Aug-09	Commissioned

63	JSW	Toranahallu	1 x 100 MW	ABB	PPB	Performance Monitoring system	Sep-05	Commissioned
64	IPGENCO	New Delhi	2 x 125 MW	ABB	Fix(DAS)	Performance Monitoring system	Jan-09	Commissioned
65	NALCO	Ankul	6 x 25 MW	ABB	---	Performance Monitoring system	Mar-08	Commissioned
66	National Fertilizer	Panipat	2 x 15 MW	ABB	PGP	Performance Monitoring system	Oct-10	Commissioned
67	National Fertilizer	Nangal	2 x 15 MW	ABB	PGP	Performance Monitoring system	May-06	Commissioned
68	National Fertilizer	Bhatinda	2 x 15 MW	ABB	PGP	Performance Monitoring system	Jan-06	Commissioned
69	Indian Oil Corporation Limited	Panipat	187 MW (5GT+1STG +1 Aux Boiler)	ABB	800XA	- Performance Monitoring System - Steam Load Optimization	Jul-10	Commissioned

YEA, Singapore = Yokogawa Singapore

YKO, South Korea = Yokogawa South Korea

YMA, Bahrain = Yokogawa Bahrain

16 Supplemental Materials

16.1 Contact Information

Company:	Kalki Communication Technologies Pvt Ltd
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Date Founded:	1998
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